JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Limits, Continuity and Differentiability

JEE Main Mathematics · Previous Year Questions with Answer Key

276	172	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	214 questions
Numerical	62 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 214 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

For the function $f(x) = e^{\sin|x|} - |x|, x \in \mathbf{R}$, consider the following statements :

Statement I : f is differentiable for all $x \in \mathbf{R}$.

Statement II : f is increasing in $\left(-\pi, -\frac{\pi}{2}\right)$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q2. JEE Main 2026 (Online) 6th April Morning Shift

The value of $\lim\limits_{x\to 0}\left(\frac{x^2\sin^2 x}{x^2-\sin^2 x}\right)$ is :

(A) 2 (B) 3 (C) 4 (D) 6

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Let $f(x) = \lim\limits_{y\to 0}\frac{(1-\cos(xy))\tan(xy)}{y^3}$. Then the number of solutions of the equation $f(x) = \sin x, x \in \mathbf{R}$ is :

(A) 0 (B) 2 (C) 3 (D) 1

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Let $f(x)$ and $g(x)$ be twice differentiable functions satisfying $f''(x) = g''(x)$ for all $x \in \mathbf{R}, f'(1) = 2g'(1) = 4$ and $g(2) = 3f(2) = 9$. Then $f(25) - g(25)$ is equal to :

(A) 20 (B) 40 (C) -20 (D) -40

Q5. JEE Main 2026 (Online) 5th April Morning Shift

The product of all possible values of α, for which

$\lim\limits_{x\to 0}\left(\frac{1-\cos(\alpha x)\cos((\alpha+1)x)\cos((\alpha+2)x)}{\sin^2((\alpha+1)x)}\right) = 2$, is :

(A) -2 (B) 1 (C) -1 (D) $\frac{5}{4}$

Q6. JEE Main 2026 (Online) 2nd April Morning Shift

If $\lim\limits_{x\to 2}\frac{\sin(x^3-5x^2+ax+b)}{\left(\sqrt{x-1}-1\right)\log_e(x-1)} = m$, then $a+b+m$ is equal to :

(A) 5 (B) 6 (C) 8 (D) 10

Q7. JEE Main 2026 (Online) 28th January Evening Shift

Let $f(x)=\lim\limits_{\theta\to 0}\left(\frac{\cos\pi x-x^{\left(\frac{2}{\theta}\right)}\sin(x-1)}{1+x^{\left(\frac{2}{\theta}\right)}(x-1)}\right),\ x\in\mathbb{R}$. Consider the following two statements :

(I) $f(x)$ is discontinuous at $x=1$.

(II) $f(x)$ is continuous at $x=-1$.

Then,

(A) Neither (I) nor (II) is True

(B) Only (II) is True

(C) Only (I) is True

(D) Both (I) and (II) are True

Q8. JEE Main 2026 (Online) 28th January Morning Shift

The value of

$\lim\limits_{x\to 0}\frac{\log_e\left(\sec(ex)\cdot\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2-e^{2\cos x}}$

is equal to

(A) $\frac{\left(e^{10}-1\right)}{2e^2\left(e^2-1\right)}$

(B) $\frac{\left(e^{20}-1\right)}{2e^2\left(e^2-1\right)}$

(C) $\frac{\left(e^{10}-1\right)}{2\left(e^2-1\right)}$

(D) $\frac{\left(e^{20}-1\right)}{2\left(e^2-1\right)}$

Q9. JEE Main 2026 (Online) 24th January Evening Shift

Let $y=y(x)$ be a differentiable function in the interval $(0,\infty)$ such that $y(1)=2$, and $\lim\limits_{t\to x}\left(\frac{t^2y(x)-x^2y(t)}{x-t}\right)=3$ for each $x>0$. Then $2y(2)$ is equal to :

(A) 27

(B) 18

(C) 23

(D) 12

Q10. JEE Main 2026 (Online) 24th January Evening Shift

Let $[t]$ denote the greatest integer less than or equal to t. If the function

$$f(x)=\begin{cases}b^2\sin\left(\frac{\pi}{2}\left[\frac{\pi}{2}(\cos x+\sin x)\cos x\right]\right), & x<0\\ \frac{\sin x-\frac{1}{2}\sin 2x}{x^3} & ,x>0\\ a & ,x=0\end{cases}$$

is continuous at $x=0$, then a^2+b^2 is equal to :

(A) $\frac{1}{2}$

(B) $\frac{5}{8}$

(C) $\frac{3}{4}$

(D) $\frac{9}{16}$

Q11. JEE Main 2026 (Online) 24th January Morning Shift

Let $\alpha,\beta\in\mathbb{R}$ be such that the function $f(x)=\begin{cases}2\alpha\left(x^2-2\right)+2\beta x & ,x<1\\ (\alpha+3)x+(\alpha-\beta) & ,x\geq 1\end{cases}$ be differentiable at all $x\in\mathbb{R}$. Then $34(\alpha+\beta)$ is equal to

(A) 48

(B) 84

(C) 36

(D) 24

Q12. JEE Main 2026 (Online) 24th January Morning Shift

If the function $f(x)=\frac{e^x(e^{\tan x-x}-1)+\log_e(\sec x+\tan x)-x}{\tan x-x}$ is continuous at $x=0$, then the value of $f(0)$ is equal to

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) 2 (D) $\frac{3}{2}$

Q13. JEE Main 2026 (Online) 23rd January Evening Shift

If $f(x)=\begin{cases}\frac{a|x|+x^2-2(\sin|x|)(\cos|x|)}{x} & ,x\neq 0\\ b & ,x=0\end{cases}$

is continuous at $x=0$, then $a+b$ is equal to :

(A) 1 (B) 2 (C) 0 (D) 4

Q14. JEE Main 2026 (Online) 23rd January Morning Shift

Let $f(x)=\begin{cases}\frac{ax^2+2ax+3}{4x^2+4x-3} & ,x\neq -\frac{3}{2},\frac{1}{2}\\ \mathrm{b} & ,x=-\frac{3}{2},\frac{1}{2}\end{cases}$ be continuous at $x=-\frac{3}{2}$. If $f\circ f(x)=\frac{7}{5}$, then x is equal to:

(A) 1.4 (B) 2 (C) 1 (D) 0

Q15. JEE Main 2026 (Online) 22nd January Evening Shift

If $\lim\limits_{x\to 0}\frac{\mathrm{e}^{(\mathrm{a}-1)x}+2\cos\,\mathrm{b}x+(\mathrm{c}-2)\mathrm{e}^{-x}}{x\cos x-\log_{\mathrm{e}}(1+x)}=2$, then $\mathrm{a}^2+\mathrm{b}^2+\mathrm{c}^2$ is equal to :

(A) 3 (B) 5 (C) 9 (D) 7

Q16. JEE Main 2026 (Online) 22nd January Evening Shift

Let $[\cdot]$ denote the greatest integer function, and let $f(x)=\min\left\{\sqrt{2}x,x^2\right\}$.

Let $\mathrm{S}=\{x\in(-2,2)$: the function $\mathrm{g}(x)=|x|\left[x^2\right]$ is discontinuous at $x\}$.

Then $\sum\limits_{x\in\mathrm{S}}f(x)$ equals

(A) $2-\sqrt{2}$ (B) $2\sqrt{6}-3\sqrt{2}$ (C) $1-\sqrt{2}$ (D) $\sqrt{6}-2\sqrt{2}$

Q17. JEE Main 2026 (Online) 21st January Morning Shift

Let $f:\mathbf{R}\to(0,\infty)$ be a twice differentiable function such that $f(3)=18, f'(3)=0$ and $f''(3)=4$.

Then $\lim\limits_{x\to 1}\left(\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right)$ is equal to :

(A) 9 (B) 18 (C) 1 (D) 2

Q18. JEE Main 2025 (Online) 8th April Evening Shift

Given below are two statements:

Statement I: $\lim\limits_{x\to 0}\left(\frac{\tan^{-1}x+\log_e\sqrt{\frac{1+x}{1-x}}-2x}{x^5}\right)=\frac{2}{5}$

Statement II: $\lim\limits_{x\to 1}\left(x^{\frac{2}{1-x}}\right)=\frac{1}{e^2}$

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q19. JEE Main 2025 (Online) 7th April Morning Shift

$\lim\limits_{x\to 0^+}\frac{\tan\left(5(x)^{\frac{1}{3}}\right)\log_e(1+3x^2)}{\left(\tan^{-1}3\sqrt{x}\right)^2\left(e^{5(x)^{\frac{4}{3}}}-1\right)}$ is equal to

(A) $\frac{5}{3}$ (B) 1 (C) $\frac{1}{3}$ (D) $\frac{1}{15}$

Q20. JEE Main 2025 (Online) 4th April Evening Shift

Let f be a differentiable function on $\mathbf{R}$ such that $f(2)=1, f'(2)=4$. Let $\lim\limits_{x\to 0}(f(2+x))^{3/x}=\mathrm{e}^{\alpha}$. Then the number of times the curve $y=4x^3-4x^2-4(\alpha-7)x-\alpha$ meets x-axis is :

(A) 3 (B) 1 (C) 2 (D) 0

Q21. JEE Main 2025 (Online) 4th April Morning Shift

Let $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $f(0)=1$ and $f(2x)-f(x)=x$ for all $x\in\mathbb{R}$. If $\lim\limits_{n\to\infty}\left\{f(x)-f\left(\frac{x}{2^n}\right)\right\}=G(x)$, then $\sum\limits_{r=1}^{10}G\left(r^2\right)$ is equal to

(A) 215 (B) 420 (C) 385 (D) 540

Q22. JEE Main 2025 (Online) 4th April Morning Shift

If $\lim\limits_{x\to 1^+}\frac{(x-1)(6+\lambda\cos(x-1))+\mu\sin(1-x)}{(x-1)^3}=-1$, where $\lambda,\mu\in\mathbb{R}$, then $\lambda+\mu$ is equal to

(A) 20 (B) 19 (C) 18 (D) 17

Q23. JEE Main 2025 (Online) 3rd April Morning Shift

Let $f(x)=\begin{cases}(1+ax)^{1/x}, & x<0\\ 1+b, & x=0\\ \frac{(x+4)^{1/2}-2}{(x+c)^{1/3}-2}, & x>0\end{cases}$ be continuous at $x=0$. Then e^abc is equal to:

(A) 64 (B) 48 (C) 36 (D) 72

Q24. JEE Main 2025 (Online) 2nd April Evening Shift

If $\lim\limits_{x\to 0}\frac{\cos(2x)+a\cos(4x)-b}{x^4}$ is finite, then $(a+b)$ is equal to:

(A) 0 (B) $\frac{3}{4}$ (C) -1 (D) $\frac{1}{2}$

Q25. JEE Main 2025 (Online) 2nd April Morning Shift

For $\alpha,\beta,\gamma\in\mathbf{R}$, if $\lim\limits_{x\to 0}\frac{x^2\sin\alpha x+(\gamma-1)e^{x^2}}{\sin 2x-\beta x}=3$, then $\beta+\gamma-\alpha$ is equal to :

(A) -1 (B) 4 (C) 6 (D) 7

Q26. JEE Main 2025 (Online) 29th January Evening Shift

Let the function $f(x)=\left(x^2-1\right)\left|x^2-ax+2\right|+\cos|x|$ be not differentiable at the two points $x=\alpha=2$ and $x=\beta$. Then the distance of the point (α,β) from the line $12x+5y+10=0$ is equal to :

(A) 5 (B) 2 (C) 4 (D) 3

Q27. JEE Main 2025 (Online) 29th January Morning Shift

The value of $\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\frac{k^3+6k^2+11k+5}{(k+3)!}\right)$ is :

(A) 5/3 (B) 2 (C) 4/3 (D) 7/3

Q28. JEE Main 2025 (Online) 24th January Evening Shift

Let $[x]$ denote the greatest integer function, and let m and n respectively be the numbers of the points, where the function $f(x)=[x]+|x-2|,-2<x<3$, is not continuous and not differentiable. Then $\mathrm{m}+\mathrm{n}$ is equal to :

(A) 6 (B) 9 (C) 8 (D) 7

Q29. JEE Main 2025 (Online) 24th January Morning Shift

$\lim\limits_{x\to 0}\operatorname{cosec}x\left(\sqrt{2\cos^2x+3\cos x}-\sqrt{\cos^2x+\sin x+4}\right)$ is:

(A) $\frac{1}{\sqrt{15}}$ (B) $\frac{1}{2\sqrt{5}}$ (C) 0 (D) $-\frac{1}{2\sqrt{5}}$

Q30. JEE Main 2025 (Online) 24th January Morning Shift

Let $f:\mathbb{R}-\{0\}\to\mathbb{R}$ be a function such that $f(x)-6f\left(\frac{1}{x}\right)=\frac{35}{3x}-\frac{5}{2}$. If the $\lim\limits_{x\to 0}\left(\frac{1}{\alpha x}+f(x)\right)=\beta;\alpha,\beta\in\mathbb{R}$, then $\alpha+2\beta$ is equal to

(A) 6 (B) 5 (C) 3 (D) 4

Q31. JEE Main 2025 (Online) 23rd January Evening Shift

$\lim\limits_{x\to\infty}\frac{(2x^2-3x+5)(3x-1)^{\frac{x}{2}}}{(3x^2+5x+4)\sqrt{(3x+2)^x}}$ is equal to :

(A) $\frac{2e}{3}$ (B) $\frac{2}{3\sqrt{e}}$ (C) $\frac{2e}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3e}}$

Q32. JEE Main 2025 (Online) 23rd January Morning Shift

If the function

$$f(x)=\begin{cases}\frac{2}{x}\{\sin(k_1+1)x+\sin(k_2-1)x\}, & x<0\\ 4, & x=0\\ \frac{2}{x}\log_e\left(\frac{2+k_1x}{2+k_2x}\right), & x>0\end{cases}$$

is continuous at $x=0$, then $k_1^2+k_2^2$ is equal to :

(A) 5 (B) 10 (C) 20 (D) 8

Q33. JEE Main 2025 (Online) 22nd January Evening Shift

If $\lim\limits_{x\to\infty}\left(\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)\right)^x=\alpha$, then the value of $\frac{\log_e\alpha}{1+\log_e\alpha}$ equals :

(A) e^{-2} (B) e^2 (C) e (D) e^{-1}

Q34. JEE Main 2025 (Online) 22nd January Morning Shift

If $\sum\limits_{r=1}^{n}T_r=\frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\left(\frac{1}{T_r}\right)$ is equal to :

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) 1 (D) 0

Q35. JEE Main 2024 (Online) 9th April Evening Shift

$\lim\limits_{x\to0}\frac{e-(1+2x)^{\frac{1}{2x}}}{x}$ is equal to

(A) $\frac{-2}{e}$ (B) $e-e^2$ (C) 0 (D) e

Q36. JEE Main 2024 (Online) 8th April Evening Shift

For $a,b>0$, let $f(x)=\begin{cases}\frac{\tan((a+1)x)+b\tan x}{x}, & x<0\\ 3, & x=0\\ \frac{\sqrt{ax+b^2x^2}-\sqrt{ax}}{b\sqrt{a}x\sqrt{x}}, & x>0\end{cases}$ be a continuous function at $x=0$. Then $\frac{b}{a}$ is equal to :

(A) 4 (B) 5 (C) 8 (D) 6

Q37. JEE Main 2024 (Online) 6th April Evening Shift

$\lim\limits_{n\to\infty}\frac{(1^2-1)(n-1)+(2^2-2)(n-2)+\cdots+((n-1)^2-(n-1))\cdot 1}{(1^3+2^3+\cdots\cdots+n^3)-(1^2+2^2+\cdots\cdots+n^2)}$ is equal to :

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) $\frac{1}{3}$

Q38. JEE Main 2024 (Online) 5th April Evening Shift

Let $f:[-1,2]\to\mathbf{R}$ be given by $f(x)=2x^2+x+\left[x^2\right]-[x]$, where $[t]$ denotes the greatest integer less than or equal to t. The number of points, where f is not continuous, is :

(A) 5 (B) 6 (C) 4 (D) 3

Q39. JEE Main 2024 (Online) 5th April Morning Shift

If the function $f(x)=\frac{\sin 3x+\alpha\sin x-\beta\cos 3x}{x^3}, x\in\mathbf{R}$, is continuous at $x=0$, then $f(0)$ is equal to :

(A) 4 (B) -2 (C) -4 (D) 2

Q40. JEE Main 2024 (Online) 4th April Evening Shift

If the function

$$f(x)=\begin{cases}\frac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}}, & x\neq 0\\ a\log_e 2\log_e 3 & ,x=0\end{cases}$$

is continuous at $x=0$, then the value of a^2 is equal to

(A) 968 (B) 1250 (C) 1152 (D) 746

Q41. JEE Main 2024 (Online) 4th April Morning Shift

Let $f:\mathbf{R}\to\mathbf{R}$ be a function given by

$$f(x)=\begin{cases}\frac{1-\cos 2x}{x^2}, & x<0\\ \alpha, & x=0,\\ \frac{\beta\sqrt{1-\cos x}}{x}, & x>0\end{cases}$$

where $\alpha,\beta\in\mathbf{R}$. If f is continuous at $x=0$, then $\alpha^2+\beta^2$ is equal to :

(A) 48 (B) 6 (C) 3 (D) 12

Q42. JEE Main 2024 (Online) 1st February Evening Shift

Let $f(x)=\left|2x^2+5|x|-3\right|, x\in\mathbf{R}$. If m and n denote the number of points where f is not continuous and not differentiable respectively, then $\mathrm{m}+\mathrm{n}$ is equal to :

(A) 5 (B) 3 (C) 2 (D) 0

Q43. JEE Main 2024 (Online) 1st February Evening Shift

Let $f(x)=\begin{cases}x-1, & x \text{ is even,}\\ 2x, & x \text{ is odd,}\end{cases} x \in \mathbf{N}$.

If for some $\mathrm{a} \in \mathbf{N}, f(f(f(\mathrm{a})))=21$, then $\lim\limits_{x \to \mathrm{a}^-}\left\{\frac{|x|^3}{\mathrm{a}}-\left[\frac{x}{\mathrm{a}}\right]\right\}$, where $[t]$ denotes the greatest integer less than or equal to t, is equal to :

(A) 169 (B) 121 (C) 225 (D) 144

Q44. JEE Main 2024 (Online) 1st February Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be defined as :

$$f(x)=\begin{cases}\frac{a-b\cos 2x}{x^2}; & x<0\\ x^2+cx+2; & 0 \le x \le 1\\ 2x+1; & x>1\end{cases}$$

If f is continuous everywhere in $\mathbf{R}$ and m is the number of points where f is NOT differentiable then $\mathrm{m}+\mathrm{a}+\mathrm{b}+\mathrm{c}$ equals :

(A) 1 (B) 4 (C) 3 (D) 2

Q45. JEE Main 2024 (Online) 31st January Evening Shift

Let $f : \mathbb{R} \to (0, \infty)$ be a strictly increasing function such that $\lim\limits_{x \to \infty}\frac{f(7x)}{f(x)}=1$. Then, the value of $\lim\limits_{x \to \infty}\left[\frac{f(5x)}{f(x)}-1\right]$ is equal to

(A) 0 (B) 4 (C) 1 (D) 7/5

Q46. JEE Main 2024 (Online) 31st January Evening Shift

Consider the function $f : (0, \infty) \to \mathbb{R}$ defined by $f(x)=e^{-|\log_e x|}$. If m and n be respectively the number of points at which f is not continuous and f is not differentiable, then $m+n$ is

(A) 0 (B) 1 (C) 2 (D) 3

Q47. JEE Main 2024 (Online) 31st January Morning Shift

$\lim\limits_{x \to 0}\frac{e^{2|\sin x|}-2|\sin x|-1}{x^2}$

(A) is equal to 1 (B) does not exist (C) is equal to -1 (D) is equal to 2

Q48. JEE Main 2024 (Online) 31st January Morning Shift

Let $g(x)$ be a linear function and $f(x)=\begin{cases} g(x) & ,x\leq 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & ,x>0 \end{cases}$, is continuous at $x=0$. If $f'(1)=f(-1)$, then the value $g(3)$ is

(A) $\log_e\left(\frac{4}{9}\right)-1$

(B) $\frac{1}{3}\log_e\left(\frac{4}{9e^{1/3}}\right)$

(C) $\log_e\left(\frac{4}{9e^{1/3}}\right)$

(D) $\frac{1}{3}\log_e\left(\frac{4}{9}\right)+1$

Q49. JEE Main 2024 (Online) 27th January Evening Shift

Consider the function $f:(0,2)\to\mathbf{R}$ defined by $f(x)=\frac{x}{2}+\frac{2}{x}$ and the function $g(x)$ defined by $g(x)=\begin{cases}\min\{f(t)\}, & 0<\mathrm{t}\leq x \text{ and } 0<x\leq 1 \\ \frac{3}{2}+x, & 1<x<2\end{cases}$. Then,

(A) g is continuous but not differentiable at $x=1$

(B) g is continuous and differentiable for all $x\in(0,2)$

(C) g is not continuous for all $x\in(0,2)$

(D) g is neither continuous nor differentiable at $x=1$

Q50. JEE Main 2024 (Online) 27th January Evening Shift

If $\lim\limits_{x\to 0}\frac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x}=\frac{1}{3}$, then $2\alpha-\beta$ is equal to :

(A) 2 (B) 1 (C) 5 (D) 7

Q51. JEE Main 2024 (Online) 27th January Morning Shift

Consider the function.

$$f(x)=\begin{cases}\frac{\mathrm{a}(7x-12-x^2)}{\mathrm{b}|x^2-7x+12|} & ,x<3 \\ 2^{\frac{\sin(x-3)}{x-[x]}} & ,x>3 \\ \mathrm{b} & ,x=3,\end{cases}$$

where $[x]$ denotes the greatest integer less than or equal to x. If S denotes the set of all ordered pairs (a, b) such that $f(x)$ is continuous at $x=3$, then the number of elements in S is :

(A) Infinitely many (B) 4 (C) 2 (D) 1

Q52. JEE Main 2024 (Online) 27th January Morning Shift

If $\mathrm{a}=\lim\limits_{x\to 0}\frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ and $\mathrm{b}=\lim\limits_{x\to 0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$, then the value of ab^3 is :

(A) 36 (B) 25 (C) 32 (D) 30

Q53. JEE Main 2023 (Online) 15th April Morning Shift

Let $[x]$ denote the greatest integer function and

$f(x)=\max\{1+x+[x], 2+x, x+2[x]\}, 0 \le x \le 2$. Let m be the number of points in $[0,2]$, where f is not continuous and n be the number of points in $(0,2)$, where f is not differentiable. Then $(m+n)^2+2$ is equal to :

(A) 3 (B) 6 (C) 2 (D) 11

Q54. JEE Main 2023 (Online) 13th April Evening Shift

If $\lim\limits_{x\to 0}\frac{e^{ax}-\cos(bx)-\frac{cxe^{-cx}}{2}}{1-\cos(2x)}=17$, then $5a^2+b^2$ is equal to

(A) 64 (B) 68 (C) 72 (D) 76

Q55. JEE Main 2023 (Online) 11th April Evening Shift

Let f and g be two functions defined by

$f(x)=\begin{cases}x+1, & x<0\\ |x-1|, & x\ge 0\end{cases}$ and $g(x)=\begin{cases}x+1, & x<0\\ 1, & x\ge 0\end{cases}$

Then $(g\circ f)(x)$ is :

(A) continuous everywhere but not differentiable at $x=1$

(B) differentiable everywhere

(C) not continuous at $x=-1$

(D) continuous everywhere but not differentiable exactly at one point

Q56. JEE Main 2023 (Online) 11th April Morning Shift

Let $f(x)=\left[x^2-x\right]+|-x+[x]|$, where $x\in\mathbb{R}$ and $[t]$ denotes the greatest integer less than or equal to t. Then, f is :

(A) continuous at $x=0$, but not continuous at $x=1$

(B) continuous at $x=0$ and $x=1$

(C) continuous at $x=1$, but not continuous at $x=0$

(D) not continuous at $x=0$ and $x=1$

Q57. JEE Main 2023 (Online) 8th April Evening Shift

If $\alpha>\beta>0$ are the roots of the equation $ax^2+bx+1=0$, and $\lim\limits_{x\to\frac{1}{\alpha}}\left(\frac{1-\cos(x^2+bx+a)}{2(1-\alpha x)^2}\right)^{\frac{1}{2}}=\frac{1}{k}\left(\frac{1}{\beta}-\frac{1}{\alpha}\right)$, then k is equal to :

(A) 2β (B) β (C) α (D) 2α

Q58. JEE Main 2023 (Online) 8th April Morning Shift

$\lim\limits_{x\to 0}\left(\left(\frac{(1-\cos^2(3x))}{\cos^3(4x)}\right)\left(\frac{\sin^3(4x)}{(\log_e(2x+1))^5}\right)\right)$ is equal to _____.

(A) 15 (B) 18 (C) 9 (D) 24

Q59. JEE Main 2023 (Online) 6th April Evening Shift

$\lim\limits_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$ is equal to :

(A) $\sqrt{2}$ (B) 1 (C) $\frac{1}{\sqrt{2}}$ (D) 0

Q60. JEE Main 2023 (Online) 6th April Morning Shift

Let $a_1, a_2, a_3, \ldots, a_n$ be n positive consecutive terms of an arithmetic progression. If $d > 0$ is its common difference, then

$\lim\limits_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+.....+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$ is

(A) $\frac{1}{\sqrt{d}}$ (B) 1 (C) 0 (D) $\sqrt{d}$

Q61. JEE Main 2023 (Online) 31st January Evening Shift

$\lim\limits_{x\to\infty}\frac{(\sqrt{3x+1}+\sqrt{3x-1})^6+(\sqrt{3x+1}-\sqrt{3x-1})^6}{\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6}x^3$

(A) is equal to 9 (B) is equal to $\frac{27}{2}$

(C) does not exist (D) is equal to 27

Q62. JEE Main 2023 (Online) 30th January Evening Shift

Let f, g and h be the real valued functions defined on $\mathbb{R}$ as

$$f(x)=\begin{cases}\frac{x}{|x|}, & x\neq 0\\ 1, & x=0\end{cases}$$

$$g(x)=\begin{cases}\frac{\sin(x+1)}{(x+1)}, & x\neq -1\\ 1, & x=-1\end{cases}$$

and $h(x)=2[x]-f(x)$, where $[x]$ is the greatest integer $\leq x$. Then the

value of $\lim\limits_{x\to 1} g(h(x-1))$ is :

(A) 1 (B) -1 (C) $\sin(1)$ (D) 0

Q63. JEE Main 2023 (Online) 29th January Morning Shift

Let $x = 2$ be a root of the equation $x^2 + px + q = 0$ and $f(x) = \begin{cases} \frac{1-\cos(x^2-4px+q^2+8q+16)}{(x-2p)^4}, & x \neq 2p \\ 0, & x = 2p \end{cases}$

Then $\lim\limits_{x \to 2p^+} [f(x)]$, where $[.]$ denotes greatest integer function, is

(A) 2 (B) 1 (C) 0 (D) -1

Q64. JEE Main 2023 (Online) 25th January Evening Shift

If the function $f(x) = \begin{cases} (1 + |\cos x|)^{\frac{\lambda}{|\cos x|}} & , \quad 0 < x < \frac{\pi}{2} \\ \mu & , \quad x = \frac{\pi}{2} \\ e^{\frac{\cot 6x}{\cot 4x}} & , \quad \frac{\pi}{2} < x < \pi \end{cases}$

is continuous at $x = \frac{\pi}{2}$, then $9\lambda + 6\log_e \mu + \mu^6 - e^{6\lambda}$ is equal to

(A) 11 (B) 10 (C) 8 (D) $2e^4 + 8$

Q65. JEE Main 2023 (Online) 25th January Morning Shift

The value of $\lim\limits_{n \to \infty} \frac{1+2-3+4+5-6+ \ldots\ldots + (3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ is :

(A) $\frac{3}{2\sqrt{2}}$ (B) $\frac{3}{2}(\sqrt{2}+1)$ (C) $3(\sqrt{2}+1)$ (D) $\frac{\sqrt{2}+1}{2}$

Q66. JEE Main 2023 (Online) 24th January Evening Shift

The set of all values of a for which $\lim\limits_{x \to a}([x-5] - [2x+2]) = 0$, where $[\alpha]$ denotes the greatest integer less than or equal to α is equal to

(A) $[-7.5, -6.5]$ (B) $(-7.5, -6.5]$ (C) $[-7.5, -6.5)$ (D) $(-7.5, -6.5)$

Q67. JEE Main 2023 (Online) 24th January Morning Shift

$\lim\limits_{t \to 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \ldots + n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ is equal to

(A) $\frac{n(n+1)}{2}$ (B) n (C) $n^2 + n$ (D) n^2

Q68. JEE Main 2023 (Online) 24th January Morning Shift

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$

Then at $x = 0$

(A) f is continuous but f' is not continuous (B) f and f' both are continuous

(C) f is continuous but not differentiable (D) f' is continuous but not differentiable

Q69. JEE Main 2022 (Online) 29th July Evening Shift

Let the function $f(x) = \begin{cases} \frac{\log_e(1+5x)-\log_e(1+\alpha x)}{x} & ; \text{ if } x \neq 0 \\ 10 & ; \text{ if } x = 0 \end{cases}$ be continuous at $x = 0$.

Then α is equal to

(A) 10 (B) -10 (C) 5 (D) -5

Q70. JEE Main 2022 (Online) 29th July Morning Shift

If $\lim\limits_{x\to 0} \frac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \frac{2}{3}$, where $\alpha, \beta, \gamma \in \mathbf{R}$, then which of the following is NOT correct?

(A) $\alpha^2 + \beta^2 + \gamma^2 = 6$ (B) $\alpha\beta + \beta\gamma + \gamma\alpha + 1 = 0$

(C) $\alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2 + 3 = 0$ (D) $\alpha^2 - \beta^2 + \gamma^2 = 4$

Q71. JEE Main 2022 (Online) 29th July Morning Shift

The number of points, where the function $f : \mathbf{R} \to \mathbf{R}$,

$f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)\left|x^2 - 5x + 4\right|$, is NOT differentiable, is :

(A) 1 (B) 2 (C) 3 (D) 4

Q72. JEE Main 2022 (Online) 28th July Evening Shift

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$f(x) = \lim\limits_{n\to\infty} \frac{\cos(2\pi x) - x^{2n}\sin(x-1)}{1 + x^{2n+1} - x^{2n}}$ is continuous for all x in :

(A) $R - \{-1\}$ (B) $\mathbb{R} - \{-1, 1\}$ (C) $R - \{1\}$ (D) $R - \{0\}$

Q73. JEE Main 2022 (Online) 27th July Evening Shift

If for $\mathrm{p} \neq \mathrm{q} \neq 0$, the function $f(x) = \frac{\sqrt[7]{\mathrm{p}(729+x)} - 3}{\sqrt[3]{729+\mathrm{q}x} - 9}$ is continuous at $x = 0$, then :

(A) $7pq\, f(0) - 1 = 0$ (B) $63q\, f(0) - \mathrm{p}^2 = 0$ (C) $21q\, f(0) - \mathrm{p}^2 = 0$ (D) $7pq\, f(0) - 9 = 0$

Q74. JEE Main 2022 (Online) 26th July Evening Shift

Let $\beta = \lim\limits_{x\to 0} \frac{\alpha x - (e^{3x} - 1)}{\alpha x (e^{3x} - 1)}$ for some $\alpha \in \mathbb{R}$. Then the value of $\alpha + \beta$ is :

(A) $\frac{14}{5}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Q75. JEE Main 2022 (Online) 26th July Morning Shift

Let f : R $\to$ R be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to :

(A) 4 (B) 10 (C) 11 (D) 16

Q76. JEE Main 2022 (Online) 26th July Morning Shift

If the function $f(x)=\begin{cases}\frac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x-\cos x} & , \quad x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)-\{0\}\\ k & , \quad x=0\end{cases}$ is continuous at x = 0, then k is equal to:

(A) 1 (B) -1 (C) e (D) 0

Q77. JEE Main 2022 (Online) 26th July Morning Shift

If $f(x)=\begin{cases}x+a & , \quad x\le 0\\ |x-4| & , \quad x>0\end{cases}$ and $g(x)=\begin{cases}x+1 & , \quad x<0\\ (x-4)^2+b & , \quad x\ge 0\end{cases}$ are continuous on R, then $(gof)(2)+(fog)(-2)$ is equal to :

(A) -10 (B) 10 (C) 8 (D) -8

Q78. JEE Main 2022 (Online) 25th July Evening Shift

$\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-(\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ is equal to

(A) 14 (B) 7 (C) $14\sqrt{2}$ (D) $7\sqrt{2}$

Q79. JEE Main 2022 (Online) 25th July Morning Shift

If $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$, then $8(\alpha+\beta)$ is equal to :

(A) 4 (B) -8 (C) -4 (D) 8

Q80. JEE Main 2022 (Online) 29th June Evening Shift

The value of $\lim\limits_{x\to 1}\frac{(x^2-1)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is equal to:

(A) $\frac{\pi^2}{6}$ (B) $\frac{\pi^2}{3}$ (C) $\frac{\pi^2}{2}$ (D) π^2

Q81. JEE Main 2022 (Online) 28th June Evening Shift

Let f, g : R $\to$ R be functions defined by

$f(x)=\begin{cases}[x] & , \quad x<0\\ |1-x| & , \quad x\ge 0\end{cases}$ and $g(x)=\begin{cases}e^x-x & , \quad x<0\\ (x-1)^2-1 & , \quad x\ge 0\end{cases}$ where [x] denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly :

(A) one point (B) two points (C) three points (D) four points

Q82. JEE Main 2022 (Online) 28th June Evening Shift

The value of

$\lim\limits_{n\to\infty} 6\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$ is equal to :

(A) 1 (B) 2 (C) 3 (D) 6

Q83. JEE Main 2022 (Online) 28th June Morning Shift

Let f : R $\to$ R be defined as

$$f(x)=\begin{bmatrix}[e^x], & x<0\\ ae^x+[x-1], & 0\le x<1\\ b+[\sin(\pi x)], & 1\le x<2\\ [e^{-x}]-c, & x\ge 2\end{bmatrix}$$

where a, b, c $\in$ R and [t] denotes greatest integer less than or equal to t. Then, which of the following statements is true?

(A) There exists a, b, c $\in$ R such that f is continuous on R.

(B) If f is discontinuous at exactly one point, then a + b + c = 1

(C) If f is discontinuous at exactly one point, then a + b + c $\neq$ 1

(D) f is discontinuous at at least two points, for any values of a, b and c

Q84. JEE Main 2022 (Online) 27th June Morning Shift

Let a be an integer such that $\lim\limits_{x\to 7}\frac{18-[1-x]}{[x-3a]}$ exists, where [t] is greatest integer $\le$ t. Then a is equal to :

(A) -6 (B) -2 (C) 2 (D) 6

Q85. JEE Main 2022 (Online) 26th June Evening Shift

$\lim\limits_{x\to 0}\frac{\cos(\sin x)-\cos x}{x^4}$ is equal to :

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{12}$

Q86. JEE Main 2022 (Online) 26th June Evening Shift

Let f(x) = min {1, 1 + x sin x}, $0\le x\le 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to

(A) (2, 0) (B) (1, 0) (C) (1, 1) (D) (2, 1)

Q87. JEE Main 2022 (Online) 26th June Morning Shift

$\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin(\cos^{-1}x)-x}{1-\tan(\cos^{-1}x)}$ is equal to :

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Q88. JEE Main 2022 (Online) 26th June Morning Shift

Let f, g : R $\to$ R be two real valued functions defined as $f(x) = \begin{cases} -|x+3| & , \quad x < 0 \\ e^x & , \quad x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x^2 + k_1 x & , \quad x < 0 \\ 4x + k_2 & , \quad x \geq 0 \end{cases}$, where k_1 and k_2 are real constants. If (gof) is differentiable at x = 0, then (gof) (-4) + (gof) (4) is equal to :

(A) $4(e^4 + 1)$ (B) $2(2e^4 + 1)$ (C) $4e^4$ (D) $2(2e^4 - 1)$

Q89. JEE Main 2022 (Online) 25th June Evening Shift

$\lim\limits_{x \to \frac{\pi}{2}} \left(\tan^2 x \left((2\sin^2 x + 3\sin x + 4)^{\frac{1}{2}} - (\sin^2 x + 6\sin x + 2)^{\frac{1}{2}}\right)\right)$ is equal to

(A) $\frac{1}{12}$ (B) $-\frac{1}{18}$ (C) $-\frac{1}{12}$ (D) $\frac{1}{6}$

Q90. JEE Main 2022 (Online) 25th June Morning Shift

Let f(x) be a polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then, the value of $\lim\limits_{x \to 1} \frac{f(x)}{x-1}$ is equal to:

(A) -15 (B) -60 (C) 60 (D) 15

Q91. JEE Main 2022 (Online) 24th June Evening Shift

Let $f(x) = \begin{cases} \frac{\sin(x-[x])}{x-[x]} & , x \in (-2,-1) \\ \max\{2x, 3[|x|]\} & , |x| < 1 \\ 1 & , otherwise \end{cases}$

where [t] denotes greatest integer $\leq$ t. If m is the number of points where f is not continuous and n is the number of points where f is not differentiable, then the ordered pair (m, n) is :

(A) (3, 3) (B) (2, 4) (C) (2, 3) (D) (3, 4)

Q92. JEE Main 2021 (Online) 31st August Evening Shift

If $\alpha = \lim\limits_{x \to \frac{\pi}{4}} \frac{\tan^3 x - \tan x}{\cos\left(x + \frac{\pi}{4}\right)}$ and $\beta = \lim\limits_{x \to 0} (\cos x)^{\cot x}$ are the roots of the equation, $ax^2 + bx - 4 = 0$, then the ordered pair (a, b) is :

(A) (1, -3) (B) (-1, 3) (C) (-1, -3) (D) (1, 3)

Q93. JEE Main 2021 (Online) 31st August Morning Shift

The function

$f(x) = \left|x^2 - 2x - 3\right| . e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly :

(A) four points (B) three points (C) two points (D) one point

Q94. JEE Main 2021 (Online) 31st August Morning Shift

If the function

$$f(x)=\begin{cases}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x<0\\ k & , \quad x=0\\ \frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1} & , \quad x>0\end{cases}$$ is continuous

at x = 0, then $\frac{1}{a}+\frac{1}{b}+\frac{4}{k}$ is equal to :

(A) -5 (B) 5 (C) -4 (D) 4

Q95. JEE Main 2021 (Online) 31st August Morning Shift

$\lim\limits_{x\to 0}\frac{\sin^2(\pi\cos^4 x)}{x^4}$ is equal to :

(A) π^2 (B) $2\pi^2$ (C) $4\pi^2$ (D) 4π

Q96. JEE Main 2021 (Online) 27th August Evening Shift

If $\lim\limits_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right)=b$, then the ordered pair (a, b) is :

(A) $\left(1,\frac{1}{2}\right)$ (B) $\left(1,-\frac{1}{2}\right)$

(C) $\left(-1,\frac{1}{2}\right)$ (D) $\left(-1,-\frac{1}{2}\right)$

Q97. JEE Main 2021 (Online) 27th August Morning Shift

If α, β are the distinct roots of $x^2 + bx + c = 0$, then

$\lim\limits_{x\to\beta}\frac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$ is equal to :

(A) $b^2 + 4c$ (B) $2(b^2 + 4c)$ (C) $2(b^2 - 4c)$ (D) $b^2 - 4c$

Q98. JEE Main 2021 (Online) 26th August Evening Shift

Let [t] denote the greatest integer less than or equal to t. Let
f(x) = x − [x], g(x) = 1 − x + [x], and h(x) = min{f(x), g(x)}, $x \in [-2, 2]$. Then h is :

(A) continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$

(B) not continuous at exactly three points in $[-2, 2]$

(C) continuous in $[-2, 2]$ but not differentiable at exactly three points in $(-2, 2)$

(D) not continuous at exactly four points in $[-2, 2]$

Q99. JEE Main 2021 (Online) 26th August Evening Shift

$\lim\limits_{x\to 2}\left(\sum\limits_{n=1}^{9}\frac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ is equal to :

(A) $\frac{9}{44}$ (B) $\frac{5}{24}$ (C) $\frac{1}{5}$ (D) $\frac{7}{36}$

Q100. JEE Main 2021 (Online) 27th July Evening Shift

The value of

$\lim\limits_{x\to 0}\left(\frac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$ is equal to :

(A) 0 (B) 4 (C) -4 (D) -1

Q101. JEE Main 2021 (Online) 27th July Evening Shift

Let $f:[0,\infty)\to[0,3]$ be a function defined by

$$f(x)=\begin{cases}\max\{\sin t:0\le t\le x\}, & 0\le x\le\pi\\ 2+\cos x, & x>\pi\end{cases}$$

Then which of the following is true?

(A) f is continuous everywhere but not differentiable exactly at one point in (0, ∞)

(B) f is differentiable everywhere in (0, ∞)

(C) f is not continuous exactly at two points in (0, ∞)

(D) f is continuous everywhere but not differentiable exactly at two points in (0, ∞)

Q102. JEE Main 2021 (Online) 27th July Morning Shift

Let $f:\left(-\frac{\pi}{4},\frac{\pi}{4}\right)\to R$ be defined as $f(x)=\begin{cases}(1+|\sin x|)^{\frac{3a}{|\sin x|}} & , \quad -\frac{\pi}{4}<x<0\\ b & , \quad x=0\\ e^{\cot 4x/\cot 2x} & , \quad 0<x<\frac{\pi}{4}\end{cases}$

If f is continuous at x = 0, then the value of $6a + b^2$ is equal to :

(A) $1-e$ (B) $e-1$ (C) $1+e$ (D) e

Q103. JEE Main 2021 (Online) 27th July Morning Shift

Let f : R $\to$ R be a function such that f(2) = 4 and f'(2) = 1. Then, the value of $\lim\limits_{x\to 2}\frac{x^2f(2)-4f(x)}{x-2}$ is equal to :

(A) 4 (B) 8 (C) 16 (D) 12

Q104. JEE Main 2021 (Online) 25th July Morning Shift

Let f : R $\rightarrow$ R be defined as

$$f(x)=\begin{cases}\frac{\lambda|x^2-5x+6|}{\mu(5x-x^2-6)}, & x<2\\ e^{\frac{\tan(x-2)}{x-[x]}}, & x>2\\ \mu, & x=2\end{cases}$$

where [x] is the greatest integer is than or equal to x. If f is continuous at x = 2, then $\lambda + \mu$ is equal to :

(A) e(−e + 1) (B) e(e − 2) (C) 1 (D) 2e − 1

Q105. JEE Main 2021 (Online) 22th July Evening Shift

Let f : R $\rightarrow$ R be defined as $f(x)=\begin{cases}\frac{x^3}{(1-\cos 2x)^2}\log_e\left(\frac{1+2xe^{-2x}}{(1-xe^{-x})^2}\right), & x\neq 0\\ \alpha, & x=0\end{cases}$

If f is continuous at x = 0, then α is equal to :

(A) 1 (B) 3 (C) 0 (D) 2

Q106. JEE Main 2021 (Online) 20th July Morning Shift

Let a function f : R $\rightarrow$ R be defined as $f(x)=\begin{cases}\sin x-e^x & if \quad x\leq 0\\ a+[-x] & if \quad 0<x<1\\ 2x-b & if \quad x\geq 1\end{cases}$

where [x] is the greatest integer less than or equal to x. If f is continuous on R, then (a + b) is equal to:

(A) 4 (B) 3 (C) 2 (D) 5

Q107. JEE Main 2021 (Online) 18th March Evening Shift

Let f : R $\rightarrow$ R be a function defined as

$$f(x)=\begin{cases}\frac{\sin(a+1)x+\sin 2x}{2x}, if\, x<0\\ b,\ if\, x=0\\ \frac{\sqrt{x+bx^3}-\sqrt{x}}{bx^{5/2}},\ if\, x>0\end{cases}$$

If f is continuous at x = 0, then the value of a + b is equal to :

(A) -3 (B) -2 (C) $-\frac{5}{2}$ (D) $-\frac{3}{2}$

Q108. JEE Main 2021 (Online) 18th March Morning Shift

If $\lim\limits_{x\to 0}\frac{\sin^{-1}x-\tan^{-1}x}{3x^3}$ is equal to L, then the value of (6L + 1) is

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) 6 (D) 2

Q109. JEE Main 2021 (Online) 18th March Morning Shift

If $f(x)=\begin{cases}\frac{1}{|x|} & ;|x|\geq 1\\ ax^2+b & ;|x|<1\end{cases}$ is differentiable at every point of the domain, then the values of a and b are respectively :

(A) $\frac{1}{2},\frac{1}{2}$ (B) $\frac{1}{2},-\frac{3}{2}$ (C) $\frac{5}{2},-\frac{3}{2}$ (D) $-\frac{1}{2},\frac{3}{2}$

Q110. JEE Main 2021 (Online) 17th March Evening Shift

The value of the limit

$\lim\limits_{\theta\to 0}\frac{\tan(\pi\cos^2\theta)}{\sin(2\pi\sin^2\theta)}$ is equal to :

(A) 0 (B) $-\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{4}$

Q111. JEE Main 2021 (Online) 17th March Evening Shift

The value of $\lim\limits_{n\to\infty}\frac{[r]+[2r]+\ldots+[nr]}{n^2}$, where r is a non-zero real number and [r] denotes the greatest integer less than or equal to r, is equal to :

(A) r (B) $\frac{r}{2}$ (C) 0 (D) 2r

Q112. JEE Main 2021 (Online) 17th March Morning Shift

The value of

$\lim\limits_{x\to 0^+}\frac{\cos^{-1}(x-[x]^2).\sin^{-1}(x-[x]^2)}{x-x^3}$, where [x] denotes the greatest integer $\leq$ x is :

(A) π (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) 0

Q113. JEE Main 2021 (Online) 16th March Evening Shift

Let f : S $\to$ S where S = (0, ∞) be a twice differentiable function such that f(x + 1) = xf(x). If g : S $\to$ R be defined as g(x) = $\log_e$ f(x), then the value of |g"(5) $-$ g"(1)| is equal to :

(A) 1 (B) $\frac{187}{144}$ (C) $\frac{197}{144}$ (D) $\frac{205}{144}$

Q114. JEE Main 2021 (Online) 16th March Evening Shift

Let $\alpha \in$ R be such that the function $f(x) = \begin{cases} \frac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}, & x \neq 0 \\ \alpha, & x = 0 \end{cases}$ is continuous at x = 0, where {x} = x $-$ [x], [x] is the greatest integer less than or equal to x. Then :

(A) no such α exists

(B) α = 0

(C) $\alpha = \frac{\pi}{4}$

(D) $\alpha = \frac{\pi}{\sqrt{2}}$

Q115. JEE Main 2021 (Online) 16th March Morning Shift

Let $S_k = \sum_{r=1}^{k} \tan^{-1}\left(\frac{6^r}{2^{2r+1}+3^{2r+1}}\right)$. Then $\lim_{k\to\infty} S_k$ is equal to :

(A) $\cot^{-1}\left(\frac{3}{2}\right)$

(B) $\frac{\pi}{2}$

(C) $\tan^{-1}(3)$

(D) $\tan^{-1}\left(\frac{3}{2}\right)$

Q116. JEE Main 2021 (Online) 16th March Morning Shift

Let the functions f : R $\to$ R and g : R $\to$ R be defined as :

$f(x) = \begin{cases} x+2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ and

$g(x) = \begin{cases} x^3, & x < 1 \\ 3x-2, & x \geq 1 \end{cases}$

Then, the number of points in R where (fog) (x) is NOT differentiable is equal to :

(A) 0

(B) 3

(C) 1

(D) 2

Q117. JEE Main 2021 (Online) 26th February Evening Shift

Let f(x) be a differentiable function at x = a with f'(a) = 2 and f(a) = 4.

Then $\lim_{x\to a} \frac{xf(a)-af(x)}{x-a}$ equals :

(A) 4 $-$ 2a

(B) 2a + 4

(C) a + 4

(D) 2a $-$ 4

Q118. JEE Main 2021 (Online) 26th February Evening Shift

Let $f(x) = \sin^{-1}x$ and $g(x) = \frac{x^2-x-2}{2x^2-x-6}$. If $g(2) = \lim_{x\to 2} g(x)$, then the domain of the function fog is :

(A) $(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$

(B) $(-\infty, -2] \cup [-1, \infty)$

(C) $(-\infty, -2] \cup \left[-\frac{3}{2}, \infty\right)$

(D) $(-\infty, -1] \cup [2, \infty)$

Q119. JEE Main 2021 (Online) 26th February Evening Shift

Let f : R $\rightarrow$ R be defined as

$$f(x)=\begin{cases}2\sin\left(-\frac{\pi x}{2}\right), if\, x<-1\\ |ax^2+x+b|,\ if-1\le x\le 1\\ \sin(\pi x),\ if\, x>1\end{cases}$$ If f(x) is continuous on R, then a + b equals :

(A) -3 (B) 3 (C) -1 (D) 1

Q120. JEE Main 2021 (Online) 26th February Morning Shift

The value of $\lim\limits_{h\to 0} 2\left\{\frac{\sqrt{3}\sin\left(\frac{\pi}{6}+h\right)-\cos\left(\frac{\pi}{6}+h\right)}{\sqrt{3}h\left(\sqrt{3}\cosh-\sinh\right)}\right\}$ is :

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{2}{\sqrt{3}}$

Q121. JEE Main 2021 (Online) 25th February Morning Shift

$\lim\limits_{n\to\infty}\left(1+\frac{1+\frac{1}{2}+.....+\frac{1}{n}}{n^2}\right)^n$ is equal to :

(A) $\frac{1}{2}$ (B) 1 (C) 0 (D) $\frac{1}{e}$

Q122. JEE Main 2021 (Online) 24th February Morning Shift

If f : R $\rightarrow$ R is a function defined by f(x)= [x - 1] $\cos\left(\frac{2x-1}{2}\right)\pi$, where [.] denotes the greatest integer function, then f is :

(A) continuous for every real x

(B) discontinuous at all integral values of x except at x = 1

(C) discontinuous only at x = 1

(D) continuous only at x = 1

Q123. JEE Main 2020 (Online) 6th September Evening Slot

Let f : R $\rightarrow$ R be a function defined by

f(x) = max {x, x^2}. Let S denote the set of all points in R, where f is not differentiable. Then S is equal to :

(A) {0, 1} (B) {0}

(C) ϕ(an empty set) (D) {1}

Q124. JEE Main 2020 (Online) 6th September Evening Slot

For all twice differentiable functions f : R $\rightarrow$ R,
with f(0) = f(1) = f'(0) = 0

(A) f"(x) $\neq$ 0, at every point x $\in$ (0, 1)

(B) f"(x) = 0, for some x $\in$ (0, 1)

(C) f"(0) = 0

(D) f"(x) = 0, at every point x $\in$ (0, 1)

Q125. JEE Main 2020 (Online) 5th September Evening Slot

$\lim\limits_{x\to 0} \frac{x\left(e^{\left(\sqrt{1+x^2+x^4}-1\right)/x}-1\right)}{\sqrt{1+x^2+x^4}-1}$

(A) is equal to 0.

(B) is equal to $\sqrt{e}$.

(C) is equal to 1.

(D) does not exist.

Q126. JEE Main 2020 (Online) 5th September Morning Slot

If the function
$f(x)=\begin{cases}k_1(x-\pi)^2-1, & x\le\pi\\ k_2\cos x, & x>\pi\end{cases}$ is
twice differentiable, then the ordered pair (k_1, k_2) is equal to :

(A) $\left(\frac{1}{2},-1\right)$ (B) (1, 1) (C) (1, 0) (D) $\left(\frac{1}{2},1\right)$

Q127. JEE Main 2020 (Online) 5th September Morning Slot

If α is positive root of the equation, p(x) = x^2 - x - 2 = 0, then

$\lim\limits_{x\to\alpha^+} \frac{\sqrt{1-\cos(p(x))}}{x+\alpha-4}$ is equal to :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{3}{\sqrt{2}}$ (D) $\frac{3}{2}$

Q128. JEE Main 2020 (Online) 4th September Evening Slot

Let $f:(0,\infty)\rightarrow(0,\infty)$ be a differentiable function such that f(1) = e and $\lim\limits_{t\to x} \frac{t^2f^2(x)-x^2f^2(t)}{t-x}=0$. If f(x) = 1, then x is equal to :

(A) $\frac{1}{e}$ (B) e (C) $\frac{1}{2e}$ (D) 2e

Q129. JEE Main 2020 (Online) 4th September Evening Slot

The function

$f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1}x, & |x| \le 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$ is :

(A) continuous on R–{–1} and differentiable on R–{–1, 1}

(B) both continuous and differentiable on R–{1}

(C) both continuous and differentiable on R–{–1}

(D) continuous on R–{1} and differentiable on R–{–1, 1}

Q130. JEE Main 2020 (Online) 3rd September Evening Slot

$\lim\limits_{x\to a} \frac{(a+2x)^{\frac{1}{3}} - (3x)^{\frac{1}{3}}}{(3a+x)^{\frac{1}{3}} - (4x)^{\frac{1}{3}}}$ $(a \neq 0)$ is equal to :

(A) $\left(\frac{2}{9}\right)\left(\frac{2}{3}\right)^{\frac{1}{3}}$ (B) $\left(\frac{2}{3}\right)\left(\frac{2}{9}\right)^{\frac{1}{3}}$

(C) $\left(\frac{2}{3}\right)^{\frac{4}{3}}$ (D) $\left(\frac{2}{9}\right)^{\frac{4}{3}}$

Q131. JEE Main 2020 (Online) 3rd September Morning Slot

Let [t] denote the greatest integer $\le$ t. If for some $\lambda \in$ R - {1, 0}, $\lim\limits_{x\to 0} \left|\frac{1-x+|x|}{\lambda - x + [x]}\right|$ = L, then L is equal to :

(A) 1 (B) 2 (C) 0 (D) $\frac{1}{2}$

Q132. JEE Main 2020 (Online) 2nd September Evening Slot

$\lim\limits_{x\to 0} \left(\tan\left(\frac{\pi}{4} + x\right)\right)^{\frac{1}{x}}$ is equal to :

(A) 2 (B) 1 (C) e (D) e^2

Q133. JEE Main 2020 (Online) 2nd September Morning Slot

If a function f(x) defined by

$$f(x) = \begin{cases} ae^x + be^{-x}, & -1 \le x < 1 \\ cx^2, & 1 \le x \le 3 \\ ax^2 + 2cx, & 3 < x \le 4 \end{cases}$$

be continuous for some a, b, c $\in$ R and f'(0) + f'(2) = e, then the value of a is :

(A) $\frac{e}{e^2-3e-13}$ (B) $\frac{1}{e^2-3e+13}$ (C) $\frac{e}{e^2-3e+13}$ (D) $\frac{e}{e^2+3e+13}$

Q134. JEE Main 2020 (Online) 9th January Evening Slot

Let [t] denote the greatest integer $\leq$ t and $\lim_{x\to 0} x\left[\frac{4}{x}\right] = A$.

Then the function, f(x) = $[x^2]\sin(\pi x)$ is discontinuous, when x is equal to :

(A) $\sqrt{A+1}$ (B) $\sqrt{A+5}$ (C) $\sqrt{A+21}$ (D) $\sqrt{A}$

Q135. JEE Main 2020 (Online) 9th January Morning Slot

Let f be any function continuous on [a, b] and twice differentiable on (a, b). If for all x $\in$ (a, b), f'(x) > 0 and f"(x) < 0, then for any c $\in$ (a, b), $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than :

(A) 1 (B) $\frac{b-c}{c-a}$ (C) $\frac{b+a}{b-a}$ (D) $\frac{c-a}{b-c}$

Q136. JEE Main 2020 (Online) 9th January Morning Slot

If $f(x) = \begin{cases} \frac{\sin(a+2)x+\sin x}{x}; & x<0 \\ b \quad ; & x=0 \\ \frac{\left(x+3x^2\right)^{\frac{1}{3}}-x^{\frac{1}{3}}}{x^{\frac{4}{3}}}; & x>0 \end{cases}$

is continuous at x = 0, then a + 2b is equal to :

(A) 0 (B) -1 (C) -2 (D) 1

Q137. JEE Main 2020 (Online) 8th January Evening Slot

Let S be the set of all functions f : [0,1] $\to$ R, which are continuous on [0,1] and differentiable on (0,1). Then for every f in S, there exists a c $\in$ (0,1), depending on f, such that

(A) $|f(c)-f(1)| < |f'(c)|$ (B) $|f(c)+f(1)| < (1+c)\,|f'(c)|$

(C) $|f(c)-f(1)| < (1-c)\,|f'(c)|$ (D) None

Q138. JEE Main 2020 (Online) 8th January Morning Slot

$\lim_{x\to 0}\left(\frac{3x^2+2}{7x^2+2}\right)^{\frac{1}{x^2}}$ is equal to

(A) e (B) e^2 (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Q139. JEE Main 2019 (Online) 12th April Evening Slot

$\lim_{x\to 0}\frac{x+2\sin x}{\sqrt{x^2+2\sin x+1}-\sqrt{\sin^2 x-x+1}}$ is :

(A) 6 (B) 1 (C) 3 (D) 2

Q140. JEE Main 2019 (Online) 12th April Evening Slot

Let f(x) = 5 − |x − 2| and g(x) = |x + 1|, x $\in$ R. If f(x) attains maximum value at α and g(x) attains minimum value at β, then $\lim_{x\to -\alpha\beta}\frac{(x-1)\left(x^2-5x+6\right)}{x^2-6x+8}$ is equal to :

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

Q141. JEE Main 2019 (Online) 12th April Morning Slot

If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim_{n\to\infty}\sum_{r=1}^{n}\alpha^r + \lim_{n\to\infty}\sum_{r=1}^{n}\beta^r$ is equal to :

(A) $\frac{7}{116}$ (B) $\frac{29}{348}$ (C) $\frac{1}{12}$ (D) $\frac{21}{346}$

Q142. JEE Main 2019 (Online) 10th April Evening Slot

If $\lim_{x\to 1}\frac{x^2-ax+b}{x-1} = 5$, then a + b is equal to :

(A) 1 (B) - 4 (C) - 7 (D) 5

Q143. JEE Main 2019 (Online) 10th April Morning Slot

If $\lim_{x\to 1}\frac{x^4-1}{x-1} = \lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$, then k is :

(A) $\frac{3}{2}$ (B) $\frac{8}{3}$ (C) $\frac{4}{3}$ (D) $\frac{3}{8}$

Q144. JEE Main 2019 (Online) 10th April Morning Slot

If $f(x) = \begin{cases} \frac{\sin(p+1)x+\sin x}{x} & , x < 0 \\ q & , x = 0 \\ \frac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & , x > 0 \end{cases}$

is continuous at x = 0, then the ordered pair (p, q) is equal to

(A) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (B) $\left(-\frac{1}{2}, \frac{3}{2}\right)$

(C) $\left(-\frac{3}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{5}{2}, \frac{1}{2}\right)$

Q145. JEE Main 2019 (Online) 10th April Morning Slot

Let f : R $\to$ R be differentiable at c $\in$ R and f(c) = 0. If g(x) = |f(x)| , then at x = c, g is :

(A) differentiable if f '(c) = 0 (B) differentiable if f '(c) $\neq$ 0

(C) not differentiable (D) not differentiable if f '(c) = 0

Q146. JEE Main 2019 (Online) 9th April Evening Slot

If the function $f(x) = \begin{cases} a|\pi - x| + 1, x \le 5 \\ b|x - \pi| + 3, x > 5 \end{cases}$

is continuous at x = 5, then the value of a − b is :-

(A) $\frac{2}{\pi-5}$ (B) $\frac{2}{5-\pi}$ (C) $\frac{-2}{\pi+5}$ (D) $\frac{2}{\pi+5}$

Q147. JEE Main 2019 (Online) 9th April Evening Slot

If $f(x) = [x] - \left[\frac{x}{4}\right]$,x $\in$ 4 , where [x] denotes the greatest integer function, then

(A) Both $\lim_{x\to 4-} f(x)$ and $\lim_{x\to 4+} f(x)$ exist but are not equal

(B) f is continuous at x = 4

(C) $\lim_{x\to 4+} f(x)$ exists but $\lim_{x\to 4-} f(x)$ does not exist

(D) $\lim_{x\to 4-} f(x)$ exists but $\lim_{x\to 4+} f(x)$ does not exist

Q148. JEE Main 2019 (Online) 9th April Morning Slot

Let ƒ(x) = 15 − |x − 10|; x $\in$ R. Then the set of all values of x, at which the function, g(x) = ƒ(ƒ(x)) is not differentiable, is :

(A) {10,15} (B) {5,10,15,20} (C) {10} (D) {5,10,15}

Q149. JEE Main 2019 (Online) 9th April Morning Slot

If the function ƒ defined on , $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by

$$f(x) = \begin{cases} \frac{\sqrt{2}\cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$$

is continuous, then k is equal to

(A) 1 (B) 1 / $\sqrt{2}$ (C) $\frac{1}{2}$ (D) 2

Q150. JEE Main 2019 (Online) 8th April Evening Slot

Let ƒ : R $\to$ R be a differentiable function satisfying ƒ'(3) + ƒ'(2) = 0.

Then $\lim_{x\to 0} \left(\frac{1+f(3+x)-f(3)}{1+f(2-x)-f(2)}\right)^{\frac{1}{x}}$ is equal to

(A) e (B) e^2 (C) e^{-1} (D) 1

Q151. JEE Main 2019 (Online) 8th April Evening Slot

Let ƒ : [−1,3] $\to$ R be defined as

$$f(x) = \begin{cases} |x| + [x] & , \quad -1 \le x < 1 \\ x + |x| & , \quad 1 \le x < 2 \\ x + [x] & , \quad 2 \le x \le 3 \end{cases}$$

where [t] denotes the greatest integer less than or equal to t. Then, ƒ is discontinuous at:

(A) only three points (B) four or more points

(C) only two points (D) only one point

Q152. JEE Main 2019 (Online) 8th April Morning Slot

$\lim\limits_{x\to 0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$ equals:

(A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$

Q153. JEE Main 2019 (Online) 12th January Evening Slot

$\lim\limits_{x\to 1^-}\frac{\sqrt{\pi}-\sqrt{2\sin^{-1}x}}{\sqrt{1-x}}$ is equal to :

(A) $\sqrt{\frac{2}{\pi}}$ (B) $\frac{1}{\sqrt{2\pi}}$ (C) $\sqrt{\frac{\pi}{2}}$ (D) $\sqrt{\pi}$

Q154. JEE Main 2019 (Online) 12th January Evening Slot

Let f be a differentiable function such that f(1) = 2 and f '(x) = f(x) for all x $\in$ R R. If h(x) = f(f(x)), then h'(1) is equal to :

(A) 4e (B) $2e^2$ (C) $4e^2$ (D) 2e

Q155. JEE Main 2019 (Online) 12th January Morning Slot

$\lim\limits_{x\to \pi/4}\frac{\cot^3 x-\tan x}{\cos\left(x+\frac{\pi}{4}\right)}$ is :

(A) $8\sqrt{2}$ (B) 4 (C) $4\sqrt{2}$ (D) 8

Q156. JEE Main 2019 (Online) 12th January Morning Slot

Let S be the set of all points in $(-\pi, \pi)$ at which the function, f(x) = min{sin x, cos x} is not differentiable. Then S is a subset of which of the following ?

(A) $\left\{-\frac{\pi}{2}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2}\right\}$

(B) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right\}$

(C) $\left\{-\frac{\pi}{4}, 0, \frac{\pi}{4}\right\}$

(D) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4}\right\}$

Q157. JEE Main 2019 (Online) 11th January Evening Slot

Let K be the set of all real values of x where the function f(x) = sin |x| − |x| + 2(x − π) cos |x| is not differentiable. Then the set K is equal to :

(A) {0, π} (B) ϕ (an empty set)

(C) { r } (D) {0}

Q158. JEE Main 2019 (Online) 11th January Evening Slot

$\lim\limits_{x\to 0}\frac{x\cot(4x)}{\sin^2 x\cot^2(2x)}$ is equal to :

(A) 0 (B) 4 (C) 1 (D) 2

Q159. JEE Main 2019 (Online) 11th January Morning Slot

Let $f(x) = \begin{cases} -1 & -2 \le x < 0 \\ x^2 - 1, & 0 \le x \le 2 \end{cases}$ and

$g(x) = |f(x)| + f(|x|)$.

Then, in the interval (−2, 2), g is :

(A) non continuous

(B) differentiable at all points

(C) not differentiable at two points

(D) not differentiable at one point

Q160. JEE Main 2019 (Online) 11th January Morning Slot

Let [x] denote the greatest integer less than or equal to x. Then $\lim\limits_{x\to 0} \frac{\tan\left(\pi\sin^2 x\right)+(|x|-\sin(x[x]))^2}{x^2}$

(A) equals π + 1

(B) equals 0

(C) does not exist

(D) equals π

Q161. JEE Main 2019 (Online) 10th January Evening Slot

Let f : $(-1, 1) \to$ R be a function defined by f(x) = max $\left\{-|x|, -\sqrt{1-x^2}\right\}$. If K be the set of all points at which f is not differentiable, then K has exactly -

(A) one element

(B) three elements

(C) five elements

(D) two elements

Q162. JEE Main 2019 (Online) 10th January Morning Slot

For each t $\in$ R , let [t] be the greatest integer less than or equal to t

Then $\lim\limits_{x\to 1^+} \frac{(1-|x|+\sin|1-x|)\sin\left(\frac{\pi}{2}[1-x]\right)}{|1-x|.[1-x]}$

(A) equals $-$ 1

(B) equals 1

(C) equals 0

(D) does not exist

Q163. JEE Main 2019 (Online) 10th January Morning Slot

Let $f(x) = \begin{cases} \max\left\{|x|, x^2\right\} & |x| \le 2 \\ 8 - 2|x| & 2 < |x| \le 4 \end{cases}$

Let S be the set of points in the interval (− 4, 4) at which f is not differentiable. Then S

(A) equals $\{-2, -1, 1, 2\}$

(B) equals $\{-2, -1, 0, 1, 2\}$

(C) equals $\{-2, 2\}$

(D) is an empty set

Q164. JEE Main 2019 (Online) 9th January Evening Slot

For each x$\in$**R**, let [x] be the greatest integer less than or equal to x.

Then $\lim\limits_{x\to 0^-} \frac{x([x]+|x|)\sin[x]}{|x|}$ is equal to :

(A) $-$ sin 1

(B) 1

(C) sin 1

(D) 0

Q165. JEE Main 2019 (Online) 9th January Morning Slot

Let f : R $\rightarrow$ R be a function defined as

$$f(x) = \begin{cases} 5 & ; \quad x \leq 1 \\ a + bx & ; \quad 1 < x < 3 \\ b + 5x & ; \quad 3 \leq x < 5 \\ 30 & ; \quad x \geq 5 \end{cases}$$

Then, f is

(A) continuous if a = 0 and b = 5

(B) continuous if a = −5 and b = 10

(C) continuous if a = 5 and b = 5

(D) not continuous for any values of a and b

Q166. JEE Main 2019 (Online) 9th January Morning Slot

$\lim\limits_{y \to 0} \frac{\sqrt{1+\sqrt{1+y^4}}-\sqrt{2}}{y^4}$

(A) exists and equals $\frac{1}{2\sqrt{2}}$

(B) exists and equals $\frac{1}{4\sqrt{2}}$

(C) exists and equals $\frac{1}{2\sqrt{2}(1+\sqrt{2})}$

(D) does not exist

Q167. JEE Main 2018 (Online) 16th April Morning Slot

$\lim\limits_{x \to 0} \frac{(27+x)^{\frac{1}{3}}-3}{9-(27+x)^{\frac{2}{3}}}$ equals.

(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) $-\frac{1}{6}$ (D) $\frac{1}{6}$

Q168. JEE Main 2018 (Online) 16th April Morning Slot

If the function f defined as

$f(x) = \frac{1}{x} - \frac{k-1}{e^{2x}-1}, x \neq 0$, is continuous at

x = 0, then the ordered pair (k, f(0)) is equal to :

(A) (3, 2) (B) (3, 1) (C) (2, 1) (D) $\left(\frac{1}{3}, 2\right)$

Q169. JEE Main 2018 (Offline)

For each t $\in R$, let [t] be the greatest integer less than or equal to t.

Then $\lim\limits_{x \to 0^+} x\left(\left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + \ldots\ldots + \left[\frac{15}{x}\right]\right)$

(A) does not exist in R

(B) is equal to 0

(C) is equal to 15

(D) is equal to 120

Q170. JEE Main 2018 (Offline)

Let S = { t $\in R$: $f(x) = |x - \pi|.\left(e^{|x|} - 1\right)\sin|x|$ is not differentiable at t}, then the set S is equal to

(A) {0, π}
(B) ϕ (an empty set)
(C) {0}
(D) {π}

Q171. JEE Main 2018 (Online) 15th April Evening Slot

$\lim\limits_{x\to 0} \frac{x\tan 2x - 2x\tan x}{(1-\cos 2x)^2}$ equals :

(A) $\frac{1}{4}$
(B) 1
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$

Q172. JEE Main 2018 (Online) 15th April Evening Slot

Let f(x) = $\begin{cases} (x-1)^{\frac{1}{2-x}}, & x > 1, x \neq 2 \\ k & , x = 2 \end{cases}$

The value of k for which f is continuous at x = 2 is :

(A) 1
(B) e
(C) e^{-1}
(D) e^{-2}

Q173. JEE Main 2018 (Online) 15th April Evening Slot

Let f(x) be a polynomial of degree 4 having extreme values at $x = 1$ and $x = 2$.

If $\lim\limits_{x\to 0}\left(\frac{f(x)}{x^2} + 1\right) = 3$ then f(-1) is equal to :

(A) $\frac{9}{2}$
(B) $\frac{5}{2}$
(C) $\frac{3}{2}$
(D) $\frac{1}{2}$

Q174. JEE Main 2018 (Online) 15th April Morning Slot

Let S = {(λ, μ) $\in$ **R** $\times$ **R** : f(t) = ($|\lambda| e^{|t|} - \mu$). sin (2|t|), **t** $\in$ **R**, is a differentiable function}. Then S is a subset of :

(A) **R** $\times$ [0, ∞)
(B) [0, ∞) $\times$ **R**
(C) **R** $\times$ ($-\infty$, 0)
(D) ($-\infty$, 0) $\times$ **R**

Q175. JEE Main 2017 (Online) 9th April Morning Slot

The value of k for which the function

$$f(x) = \begin{cases} \left(\frac{4}{5}\right)^{\frac{\tan 4x}{\tan 5x}}, & 0 < x < \frac{\pi}{2} \\ k + \frac{2}{5}, & x = \frac{\pi}{2} \end{cases}$$

is continuous at x = $\frac{\pi}{2}$, is :

(A) $\frac{17}{20}$
(B) $\frac{2}{5}$
(C) $\frac{3}{5}$
(D) $-\frac{2}{5}$

Q176. JEE Main 2017 (Online) 8th April Morning Slot

$\lim\limits_{x\to 3}\frac{\sqrt{3x}-3}{\sqrt{2x-4}-\sqrt{2}}$ is equal to :

(A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2\sqrt{2}}$

Q177. JEE Main 2017 (Offline)

$\lim\limits_{x\to\frac{\pi}{2}}\frac{\cot x-\cos x}{(\pi-2x)^3}$ equals

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{24}$

Q178. JEE Main 2016 (Online) 10th April Morning Slot

$\lim\limits_{x\to 0}\frac{(1-\cos 2x)^2}{2x\tan x-x\tan 2x}$ is :

(A) -2 (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) 2

Q179. JEE Main 2016 (Online) 10th April Morning Slot

Let a, b $\in$ **R**, (a $\neq$ 0). If the function *f* defined as

$$f(x)=\begin{cases}\frac{2x^2}{a}, & 0\le x<1\\ a, & 1\le x<\sqrt{2}\\ \frac{2b^2-4b}{x^3}, & \sqrt{2}\le x<\infty\end{cases}$$

is continuous in the interval [0, ∞), then an ordered pair (a, b) is :

(A) $\left(\sqrt{2},1-\sqrt{3}\right)$ (B) $\left(-\sqrt{2},1+\sqrt{3}\right)$

(C) $\left(\sqrt{2},-1+\sqrt{3}\right)$ (D) $\left(-\sqrt{2},1-\sqrt{3}\right)$

Q180. JEE Main 2016 (Online) 9th April Morning Slot

If $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}-\frac{4}{x^2}\right)^{2x}=e^3$, then 'a' is equal to :

(A) 2 (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Q181. JEE Main 2016 (Online) 9th April Morning Slot

If the function

$$f(x) = \begin{cases} -x & x < 1 \\ a + \cos^{-1}(x+b), & 1 \le x \le 2 \end{cases}$$

is differentiable at x = 1, then $\frac{a}{b}$ is equal to :

(A) $\frac{\pi-2}{2}$ (B) $\frac{-\pi-2}{2}$

(C) $\frac{\pi+2}{2}$ (D) $-1-\cos^{-1}(2)$

Q182. JEE Main 2016 (Offline)

For $x \in R,\ f(x) = |\log 2 - \sin x|$

and $g(x) = f(f(x)),$ then :

(A) g is not differentiable at $x = 0$

(B) $g'(0) = \cos(\log 2)$

(C) $g'(0) = -\cos(\log 2)$

(D) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$

Q183. JEE Main 2016 (Offline)

Let $p = \lim\limits_{x\to 0^+}\left(1+\tan^2\sqrt{x}\right)^{\frac{1}{2x}}$ then $\log p$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) 2 (D) 1

Q184. JEE Main 2015 (Offline)

$\lim\limits_{x\to 0}\frac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$ is equal to

(A) 2 (B) $\frac{1}{2}$ (C) 4 (D) 3

Q185. JEE Main 2015 (Offline)

If the function.

$$g(x) = \begin{cases} k\sqrt{x+1}, & 0 \le x \le 3 \\ mx+2, & 3 < x \le 5 \end{cases}$$

is differentiable, then the value of $k+m$ is :

(A) $\frac{10}{3}$ (B) 4 (C) 2 (D) $\frac{16}{5}$

Q186. JEE Main 2014 (Offline)

$\lim_{x\to 0}\frac{\sin\left(\pi\cos^2 x\right)}{x^2}$ is equal to :

(A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) 1

Q187. JEE Main 2013 (Offline)

$\lim_{x\to 0}\frac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$ is equal to

(A) $-\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

Q188. AIEEE 2012

If $f : R \to R$ is a function defined by

$f(x) = [x]\cos\left(\frac{2x-1}{2}\right)\pi$,

where [x] denotes the greatest integer function, then f is

(A) continuous for every real x

(B) discontinuous only at $x = 0$

(C) discontinuous only at non-zero integral values of x

(D) continuous only at $x = 0$

Q189. AIEEE 2012

Consider the function, $f(x) = |x-2| + |x-5|, x \in R$

Statement - 1 : $f'(4) = 0$

Statement - 2 : f is continuous in [2, 5], differentiable in (2, 5) and $f(2) = f(5)$

(A) Statement - 1 is false, statement - 2 is true

(B) Statement - 1 is true, statement - 2 is true; statement - 2 is a correct explanation for statement - 1

(C) Statement - 1 is true, statement - 2 is true; statement - 2 is not a correct explanation for statement - 1

(D) Statement - 1 is true, statement - 2 is false

Q190. AIEEE 2011

$\lim_{x\to 2}\left(\frac{\sqrt{1-\cos\{2(x-2)\}}}{x-2}\right)$

(A) Equals $\sqrt{2}$ (B) Equals $-\sqrt{2}$

(C) Equals $\frac{1}{\sqrt{2}}$ (D) does not exist

Q191. AIEEE 2011

The value of p and q for which the function

$$f(x)=\begin{cases}\frac{\sin(p+1)x+\sin x}{x} & ,x<0\\ q & ,x=0\\ \frac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & ,x>0\end{cases}$$

is continuous for all x in R, are

(A) $p=\frac{5}{2}, q=\frac{1}{2}$ (B) $p=-\frac{3}{2}, q=\frac{1}{2}$

(C) $p=\frac{1}{2}, q=\frac{3}{2}$ (D) $p=\frac{1}{2}, q=-\frac{3}{2}$

Q192. AIEEE 2010

Let $f: R \to R$ be a positive increasing function with

$\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1$. Then $\lim_{x\to\infty}\frac{f(2x)}{f(x)}=$

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 3 (D) 1

Q193. AIEEE 2009

Let $f(x)=x|x|$ and $g(x)=\sin x$.

Statement-1: gof is differentiable at $x=0$ and its derivative is continuous at that point.

Statement-2: gof is twice differentiable at $x=0$.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(B) Statement-1 is true, Statement-2 is false

(C) Statement-1 is false, Statement-2 is true

(D) Statement-1 is true, Statement-2 is true Statement-2 is a correct explanation for Statement-1

Q194. AIEEE 2008

Let $f(x)=\begin{cases}(x-1)\sin\frac{1}{x-1} & if\, x\neq 1\\ 0 & if\, x=1\end{cases}$

Then which one of the following is true?

(A) f is neither differentiable at x = 0 nor at x = 1

(B) f is differentiable at x = 0 and at x = 1

(C) f is differentiable at x = 0 but not at x = 1

(D) f is differentiable at x = 1 but not at x = 0

Q195. AIEEE 2007

Let $f : R \to R$ be a function defined by

$f(x) = \min\{x + 1, |x| + 1\}$, then which of the following is true?

(A) $f(x)$ is differentiale everywhere

(B) $f(x)$ is not differentiable at x = 0

(C) $f(x) > 1$ for all $x \in R$

(D) $f(x)$ is not differentiable at x = 1

Q196. AIEEE 2007

The function $f : R/\{0\} \to R$ given by

$f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$

can be made continuous at x = 0 by defining $f(0)$ as

(A) 0

(B) 1

(C) 2

(D) -1

Q197. AIEEE 2006

The set of points where $f(x) = \frac{x}{1+|x|}$ is differentiable is

(A) $(-\infty, 0) \cup (0, \infty)$

(B) $(-\infty, -1) \cup (-1, \infty)$

(C) $(-\infty, \infty)$

(D) $(0, \infty)$

Q198. AIEEE 2005

Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then

$\lim_{x\to\alpha} \frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$ is equal to

(A) $\frac{a^2(\alpha-\beta)^2}{2}$

(B) 0

(C) $-\frac{a^2(\alpha-\beta)^2}{2}$

(D) $\frac{(\alpha-\beta)^2}{2}$

Q199. AIEEE 2005

Suppose $f(x)$ is differentiable at x = 1 and

$\lim_{h\to 0} \frac{1}{h} f(1 + h) = 5$, then $f'(1)$ equals

(A) 3

(B) 4

(C) 5

(D) 6

Q200. AIEEE 2005

If f is a real valued differentiable function satisfying

$|f(x) - f(y)| \leq (x-y)^2, x, y \in R$
and $f(0) = 0$, then $f(1)$ equals

(A) -1 (B) 0 (C) 2 (D) 1

Q201. AIEEE 2004

If $\lim_{x\to\infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, then the value of a and b, are

(A) a = 1 and b = 2 (B) a = 1 and $b \in R$ (C) $a \in R$ and b = 2 (D) $a \in R$ and $b \in R$

Q202. AIEEE 2004

Let $f(x) = \frac{1-\tan x}{4x-\pi}$, $x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$.

If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is

(A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 1

Q203. AIEEE 2003

If $\lim_{x\to 0} \frac{\log(3+x)-\log(3-x)}{x}$ = k, the value of k is

(A) $-\frac{2}{3}$ (B) 0 (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

Q204. AIEEE 2003

Let $f(a) = g(a) = k$ and their nth derivatives
$f^n(a), g^n(a)$ exist and are not equal for some n. Further if

$\lim_{x\to a} \frac{f(a)g(x)-f(a)-g(a)f(x)+f(a)}{g(x)-f(x)} = 4$

then the value of k is

(A) 0 (B) 4 (C) 2 (D) 1

Q205. AIEEE 2003

$\lim_{x\to\frac{\pi}{2}} \frac{\left[1-\tan\left(\frac{x}{2}\right)\right][1-\sin x]}{\left[1+\tan\left(\frac{x}{2}\right)\right][\pi-2x]^3}$ is

(A) ∞ (B) $\frac{1}{8}$ (C) 0 (D) $\frac{1}{32}$

Q206. AIEEE 2003

If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$

then $f(x)$ is

(A) discontinuous everywhere

(B) continuous as well as differentiable for all x

(C) continuous for all x but not differentiable at x = 0

(D) neither differentiable nor continuous at x = 0

Q207. AIEEE 2002

$\lim_{x\to 0} \frac{\sqrt{1-\cos 2x}}{\sqrt{2}x}$ is

(A) 1 (B) -1 (C) zero (D) does not exist

Q208. AIEEE 2002

$\lim_{x\to\infty} \left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$

(A) e^4 (B) e^2 (C) e^3 (D) 1

Q209. AIEEE 2002

Let $f(2) = 4$ and $f'(x) = 4$.

Then $\lim_{x\to 2} \frac{xf(2)-2f(x)}{x-2}$ is given by

(A) 2 (B) -2 (C) -4 (D) 3

Q210. AIEEE 2002

$\lim_{x\to 0} \frac{\log x^n - [x]}{[x]}$, $n \in N$, ([x] denotes the greatest integer less than or equal to x)

(A) has value -1 (B) has value 0 (C) has value 1 (D) does not exist

Q211. AIEEE 2002

If $f(1) = 1, f'(1) = 2$, then

$\lim_{x\to 1} \frac{\sqrt{f(x)}-1}{\sqrt{x}-1}$ is

(A) 2 (B) 4 (C) 1 (D) $\frac{1}{2}$

Q212. AIEEE 2002

f is defined in $[-5, 5]$ as

$f(x) = x$ if x is rational

$= -x$ if x is irrational. Then

(A) $f(x)$ is continuous at every x, except $x = 0$

(B) $f(x)$ is discontinuous at every x, except $x = 0$

(C) $f(x)$ is continuous everywhere

(D) $f(x)$ is discontinuous everywhere

Q213. AIEEE 2002

f(x) and g(x) are two differentiable functions on [0, 2] such that

f"(x) - g"(x) = 0, f'(1) = 2, g'(1) = 4, f(2) = 3, g(2) = 9

then f(x) - g(x) at x = $\frac{3}{2}$ is

(A) 0 (B) 2 (C) 10 (D) -5

Q214. AIEEE 2002

If f(x + y) = f(x).f(y) $\forall$ x, y and f(5) = 2, f'(0) = 3, then
f'(5) is

(A) 0 (B) 1 (C) 6 (D) 2

Numerical 62 questions

Q215. JEE Main 2026 (Online) 6th April Evening Shift

Let $f(x) = \begin{cases} x^3 + 8; & x < 0, \\ x^2 - 4; & x \geq 0, \end{cases}$ and $g(x) = \begin{cases} (x-8)^{1/3}; & x < 0, \\ (x+4)^{1/2}; & x \geq 0. \end{cases}$

Then the number of points, where the function $g \circ f$ is discontinuous, is ____ .

Q216. JEE Main 2026 (Online) 4th April Evening Shift

Let $f(x) = \begin{cases} e^{x-1} & , x < 0 \\ x^2 - 5x + 6 & , x \geq 0 \end{cases}$ and $g(x) = f(|x|) + |f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha + \beta$ is equal to ____

Q217. JEE Main 2026 (Online) 4th April Morning Shift

The number of points, at which the function
$f(x)=\max\left\{6x, 2+3x^2\right\}+|x-1|\cos\left|x^2-\frac{1}{4}\right|, x\in(-\pi,\pi)$, is not differentiable, is

____ .

Q218. JEE Main 2026 (Online) 2nd April Evening Shift

The number of points in the interval $[2,4]$, at which the function $f(x)=\left[x^2-x-\frac{1}{2}\right]$, where $[\cdot]$ denotes the greatest integer function, is discontinuous, is _____.

Q219. JEE Main 2025 (Online) 7th April Evening Shift

If the function $f(x)=\frac{\tan(\tan x)-\sin(\sin x)}{\tan x-\sin x}$ is continuous at $x=0$, then $f(0)$ is equal to _____.

Q220. JEE Main 2025 (Online) 7th April Evening Shift

For $\mathrm{t}>-1$, let α_{t} and β_{t} be the roots of the equation

$\left((\mathrm{t}+2)^{1/7}-1\right)x^2+\left((\mathrm{t}+2)^{1/6}-1\right)x+\left((\mathrm{t}+2)^{1/21}-1\right)=0$

If $\lim\limits_{\mathrm{t}\to-1^+}\alpha_{\mathrm{t}}=\mathrm{a}$ and $\lim\limits_{\mathrm{t}\to-1^+}\beta_{\mathrm{t}}=\mathrm{b}$, then $72(a+b)^2$ is equal to _____.

Q221. JEE Main 2025 (Online) 7th April Morning Shift

The number of points of discontinuity of the function $f(x)=\left[\frac{x^2}{2}\right]-[\sqrt{x}], x\in[0,4]$, where $[\cdot]$ denotes the greatest integer function, is _____.

Q222. JEE Main 2025 (Online) 4th April Morning Shift

Let m and n be the number of points at which the function $f(x)=\max\left\{x, x^3, x^5, \ldots x^{21}\right\}, x\in\mathbb{R}$, is not differentiable and not continuous, respectively. Then $m+n$ is equal to _____.

Q223. JEE Main 2025 (Online) 3rd April Evening Shift

If $\lim\limits_{x\to0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}}=p$, then $96\log_{\mathrm{e}}p$ is equal to _____

Q224. JEE Main 2025 (Online) 29th January Morning Shift

Let [t] be the greatest integer less than or equal to t. Then the least value of p ∈ N for which

$\lim\limits_{x\to0^+}\left(x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+\ldots+\left[\frac{p}{x}\right]\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+\ldots+\left[\frac{9^2}{x^2}\right]\right)\right)\geq1$ is equal to _____.

Q225. JEE Main 2025 (Online) 28th January Evening Shift

Let $f(x)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\left(\frac{\tan\left(x/2^{r+1}\right)+\tan^3\left(x/2^{r+1}\right)}{1-\tan^2\left(x/2^{r+1}\right)}\right)$. Then $\lim\limits_{x\to 0}\frac{e^x-e^{f(x)}}{(x-f(x))}$ is equal to _____.

Q226. JEE Main 2025 (Online) 28th January Morning Shift

Let $\mathrm{f}(x)=\begin{cases}3x, & x<0\\ \min\{1+x+[x], x+2[x]\}, & 0\le x\le 2\\ 5, & x>2\end{cases}$

where [.] denotes greatest integer function. If α and β are the number of points, where f is not continuous and is not differentiable, respectively, then $\alpha+\beta$ equals _____ .

Q227. JEE Main 2025 (Online) 22nd January Morning Shift

Let the function,

$f(x)=\begin{cases}-3\mathrm{a}x^2-2, & x<1\\ \mathrm{a}^2+\mathrm{b}x, & x\geqslant 1\end{cases}$

be differentiable for all $x\in\mathbf{R}$, where $\mathrm{a}>1,\ \mathrm{b}\in\mathbf{R}$. If the area of the region enclosed by $y=f(x)$ and the line $y=-20$ is $\alpha+\beta\sqrt{3}, \alpha,\beta\in Z$, then the value of $\alpha+\beta$ is _____ .

Q228. JEE Main 2024 (Online) 9th April Morning Shift

Let $f:(0,\pi)\to\mathbf{R}$ be a function given by $f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ \mathrm{a}-8, & x=\frac{\pi}{2}\\ (1+|\cot x|)^{\frac{\mathrm{b}}{\mathrm{a}}|\tan x|}, & \frac{\pi}{2}<x<\pi\end{cases}$

where $\mathrm{a},\mathrm{b}\in\mathbf{Z}$. If f is continuous at $x=\frac{\pi}{2}$, then $\mathrm{a}^2+\mathrm{b}^2$ is equal to _____.

Q229. JEE Main 2024 (Online) 8th April Evening Shift

If $\alpha=\lim\limits_{x\to 0^+}\left(\frac{\mathrm{e}^{\sqrt{\tan x}}-\mathrm{e}^{\sqrt{x}}}{\sqrt{\tan x}-\sqrt{x}}\right)$ and $\beta=\lim\limits_{x\to 0}(1+\sin x)^{\frac{1}{2}\cot x}$ are the roots of the quadratic equation $\mathrm{a}x^2+\mathrm{b}x-\sqrt{\mathrm{e}}=0$, then $12\log_{\mathrm{e}}(\mathrm{a}+\mathrm{b})$ is equal to _____.

Q230. JEE Main 2024 (Online) 8th April Morning Shift

The value of $\lim\limits_{x\to 0}2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}\ldots\ldots\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____.

Q231. JEE Main 2024 (Online) 6th April Evening Shift

Let $[t]$ denote the greatest integer less than or equal to t. Let $f:[0,\infty)\to\mathbf{R}$ be a function defined by $f(x)=\left[\frac{x}{2}+3\right]-[\sqrt{x}]$. Let S be the set of all points in the interval $[0,8]$ at which f is not continuous. Then $\sum\limits_{a\in S}a$ is equal to _____.

Q232. JEE Main 2024 (Online) 5th April Evening Shift

Let $\mathrm{a} > 0$ be a root of the equation $2x^2 + x - 2 = 0$. If $\lim\limits_{x \to \frac{1}{a}} \frac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{(1-ax)^2} = \alpha + \beta\sqrt{17}$, where $\alpha, \beta \in Z$, then $\alpha + \beta$ is equal to _____.

Q233. JEE Main 2024 (Online) 5th April Morning Shift

Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $2f(2) + 3f(3)$ is equal to _____.

Q234. JEE Main 2024 (Online) 4th April Morning Shift

If $\lim\limits_{x \to 1} \frac{(5x+1)^{1/3} - (x+5)^{1/3}}{(2x+3)^{1/2} - (x+4)^{1/2}} = \frac{\mathrm{m}\sqrt{5}}{\mathrm{n}(2\mathrm{n})^{2/3}}$, where $\gcd(\mathrm{m}, \mathrm{n}) = 1$, then $8\,\mathrm{m} + 12\mathrm{n}$ is equal to _____.

Q235. JEE Main 2024 (Online) 1st February Morning Shift

Let $\{x\}$ denote the fractional part of x and $f(x) = \frac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}$, $x \neq 0$. If L and R respectively denotes the left hand limit and the right hand limit of $f(x)$ at $x = 0$, then $\frac{32}{\pi^2}\left(\mathrm{L}^2 + \mathrm{R}^2\right)$ is equal to _____.

Q236. JEE Main 2024 (Online) 31st January Evening Shift

If $\lim\limits_{x \to 0} \frac{ax^2 e^x - b\log_e(1+x) + cxe^{-x}}{x^2 \sin x} = 1$, then $16\left(a^2 + b^2 + c^2\right)$ is equal to _____.

Q237. JEE Main 2024 (Online) 30th January Morning Shift

If the function

$$f(x) = \begin{cases} \frac{1}{|x|}, & |x| \geqslant 2 \\ \mathrm{a}x^2 + 2\,\mathrm{b}, & |x| < 2 \end{cases}$$

is differentiable on $\mathbf{R}$, then $48(a + b)$ is equal to _____.

Q238. JEE Main 2024 (Online) 29th January Evening Shift

Let $f(x) = \sqrt{\lim\limits_{r \to x} \left\{ \frac{2r^2[(f(r))^2 - f(x)f(r)]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$ be differentiable in $(-\infty, 0) \cup (0, \infty)$ and $f(1) = 1$. Then the value of ea, such that $f(a) = 0$, is equal to _____.

Q239. JEE Main 2023 (Online) 12th April Morning Shift

Let $[x]$ be the greatest integer $\leq x$. Then the number of points in the interval $(-2, 1)$, where the function $f(x) = |[x]| + \sqrt{x - [x]}$ is discontinuous, is _____.

Q240. JEE Main 2023 (Online) 10th April Morning Shift

Let $f : (-2, 2) \to R$ be defined by $f(x) = \begin{cases} x[x], & -2 < x < 0 \\ (x-1)[x], & 0 \leq x < 2 \end{cases}$ where $[x]$ denotes the greatest integer function. If m and n respectively are the number of points in $(-2, 2)$ at which $y = |f(x)|$ is not continuous and not differentiable, then $m + n$ is equal to _____.

Q241. JEE Main 2023 (Online) 8th April Evening Shift

Let k and m be positive real numbers such that the function $f(x) = \begin{cases} 3x^2 + k\sqrt{x+1}, & 0 < x < 1 \\ mx^2 + k^2, & x \geq 1 \end{cases}$ is differentiable for all $x > 0$. Then $\frac{8f'(8)}{f'(\frac{1}{8})}$ is equal to _____.

Q242. JEE Main 2023 (Online) 6th April Morning Shift

Let $a \in \mathbb{Z}$ and $[\mathrm{t}]$ be the greatest integer $\leq \mathrm{t}$. Then the number of points, where the function $f(x) = [a + 13\sin x], x \in (0, \pi)$ is not differentiable, is _____.

Q243. JEE Main 2022 (Online) 29th July Evening Shift

If $[t]$ denotes the greatest integer $\leq t$, then the number of points, at which the function $f(x) = 4|2x + 3| + 9\left[x + \frac{1}{2}\right] - 12[x + 20]$ is not differentiable in the open interval $(-20, 20)$, is _____.

Q244. JEE Main 2022 (Online) 28th July Morning Shift

Let $f : [0, 1] \to \mathbf{R}$ be a twice differentiable function in $(0, 1)$ such that $f(0) = 3$ and $f(1) = 5$. If the line $y = 2x + 3$ intersects the graph of f at only two distinct points in $(0, 1)$, then the least number of points $x \in (0, 1)$, at which $f''(x) = 0$, is _____.

Q245. JEE Main 2022 (Online) 28th July Morning Shift

$\lim\limits_{x \to 0} \left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)} \right)^{\frac{100}{x}}$ is equal to _____.

Q246. JEE Main 2022 (Online) 25th July Morning Shift

Let $f(x) = \begin{cases} |4x^2 - 8x + 5|, & \text{if } 8x^2 - 6x + 1 \geqslant 0 \\ [4x^2 - 8x + 5], & \text{if } 8x^2 - 6x + 1 < 0, \end{cases}$ where $[\alpha]$ denotes the greatest integer less than or equal to α. Then the number of points in $\mathbf{R}$ where f is not differentiable is _____.

Q247. JEE Main 2022 (Online) 30th June Morning Shift

Suppose $\lim\limits_{x\to 0}\frac{F(x)}{x^3}$ exists and is equal to L, where

$$F(x)=\begin{vmatrix} a+\sin\frac{x}{2} & -b\cos x & 0 \\ -b\cos x & 0 & a+\sin\frac{x}{2} \\ 0 & a+\sin\frac{x}{2} & -b\cos x \end{vmatrix}.$$

Then, -112 L is equal to _____.

Q248. JEE Main 2022 (Online) 28th June Evening Shift

If $\lim\limits_{x\to 1}\frac{\sin(3x^2-4x+1)-x^2+1}{2x^3-7x^2+ax+b}=-2$, then the value of (a $-$ b) is equal to _____.

Q249. JEE Main 2022 (Online) 27th June Evening Shift

Let [t] denote the greatest integer $\le$ t and {t} denote the fractional part of t. The integral value of α for which the left hand limit of the function

$f(x)=[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$ at x = 0 is equal to $\alpha-\frac{4}{3}$, is _____.

Q250. JEE Main 2022 (Online) 25th June Evening Shift

Let $f(x)=\left[2x^2+1\right]$ and $g(x)=\begin{cases}2x-3, & x<0\\ 2x+3, & x\ge 0\end{cases}$, where [t] is the greatest integer $\le$ t. Then, in the open interval $(-1, 1)$, the number of points where fog is discontinuous is equal to _____.

Q251. JEE Main 2022 (Online) 24th June Morning Shift

The number of points where the function

$$f(x)=\begin{cases}|2x^2-3x-7| & if \quad x\le -1\\ [4x^2-1] & if \quad -1<x<1\\ |x+1|+|x-2| & if \quad x\ge 1\end{cases}$$

[t] denotes the greatest integer $\le$ t, is discontinuous is _____.

Q252. JEE Main 2021 (Online) 1st September Evening Shift

Let $f(x)=x^6+2x^4+x^3+2x+3$, x $\in$ R. Then the natural number n for which $\lim\limits_{x\to 1}\frac{x^nf(1)-f(x)}{x-1}=44$ is _____.

Q253. JEE Main 2021 (Online) 1st September Evening Shift

Let [t] denote the greatest integer $\le$ t. The number of points where the function $f(x)=[x]\left|x^2-1\right|+\sin\left(\frac{\pi}{[x]+3}\right)-[x+1], x\in(-2,2)$ is not continuous is _____.

Q254. JEE Main 2021 (Online) 26th August Morning Shift

Let a, b $\in$ R, b $\neq$ 0, Define a function

$$f(x) = \begin{cases} a\sin\frac{\pi}{2}(x-1), & for\, x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & for\, x > 0 \end{cases}.$$

If f is continuous at x = 0, then $10 - ab$ is equal to _____.

Q255. JEE Main 2021 (Online) 27th July Morning Shift

Let $f : [0, 3] \to R$ be defined by $f(x) = \min\{x - [x], 1 + [x] - x\}$ where [x] is the greatest integer less than or equal to x. Let P denote the set containing all x $\in$ [0, 3] where f i discontinuous, and Q denote the set containing all x $\in$ (0, 3) where f is not differentiable. Then the sum of number of elements in P and Q is equal to _____.

Q256. JEE Main 2021 (Online) 25th July Evening Shift

Consider the function

where P(x) is a polynomial such that P" (x) is always a constant and P(3) = 9. If f(x) is continuous at x = 2, then P(5) is equal to _____.

$$f(x) = \begin{cases} \dfrac{P(x)}{\sin(x-2)}, & x \neq 2 \\ 7, & x = 2 \end{cases}$$

Q257. JEE Main 2021 (Online) 22th July Evening Shift

Let f : R $\to$ R be a function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & if \quad |x| \leq 2 \\ 0 & if \quad |x| > 2 \end{cases}$

Let g : R $\to$ R be given by $g(x) = f(x + 2) - f(x - 2)$. If n and m denote the number of points in R where g is not continuous and not differentiable, respectively, then n + m is equal to _____.

Q258. JEE Main 2021 (Online) 20th July Evening Shift

Let a function g : [0, 4] $\rightarrow$ R be defined as

$g(x)=\begin{cases}\max\limits_{0\le t\le x}\{t^3-6t^2+9t-3\}, & 0\le x\le 3\\ 4-x, & 3<x\le 4\end{cases}$, then the number of points in the interval (0, 4) where g(x) is NOT differentiable, is _____.

Q259. JEE Main 2021 (Online) 20th July Evening Shift

If $\lim\limits_{x\to 0}\frac{\alpha x e^x-\beta\log_e(1+x)+\gamma x^2e^{-x}}{x\sin^2 x}=10, \alpha,\beta,\gamma\in R$, then the value of $\alpha+\beta+\gamma$ is _____.

Q260. JEE Main 2021 (Online) 20th July Morning Shift

If the value of $\lim\limits_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\left(\frac{x+2}{x^2}\right)}$ is equal to e^a, then a is equal to _____.

Q261. JEE Main 2021 (Online) 18th March Evening Shift

Let f : R $\rightarrow$ R satisfy the equation f(x + y) = f(x) . f(y) for all x, y $\in$R and f(x) $\neq$ 0 for any x$\in$R. If the function f is differentiable at x = 0 and f'(0) = 3, then

$\lim\limits_{h\to 0}\frac{1}{h}(f(h)-1)$ is equal to _____.

Q262. JEE Main 2021 (Online) 17th March Morning Shift

If the function $f(x)=\frac{\cos(\sin x)-\cos x}{x^4}$ is continuous at each point in its domain and $f(0)=\frac{1}{k}$, then k is _____.

Q263. JEE Main 2021 (Online) 16th March Evening Shift

Let f : R $\rightarrow$ R and g : R $\rightarrow$ R be defined as

$f(x)=\begin{cases}x+a, & x<0\\ |x-1|, & x\ge 0\end{cases}$ and

$g(x)=\begin{cases}x+1, & x<0\\ (x-1)^2+b, & x\ge 0\end{cases}$,

where a, b are non-negative real numbers. If (gof) (x) is continuous for all x $\in$ R, then a + b is equal to _____.

Q264. JEE Main 2021 (Online) 16th March Morning Shift

If $\lim\limits_{x\to 0}\frac{ae^x-b\cos x+ce^{-x}}{x\sin x}=2$, then a + b + c is equal to _____.

Q265. JEE Main 2021 (Online) 25th February Evening Shift

A function f is defined on $[-3, 3]$ as

$f(x)=\begin{cases}\min\{|x|,2-x^2\}, & -2\le x\le 2\\ [|x|], & 2<|x|\le 3\end{cases}$ where [x] denotes the greatest integer $\le$ x. The number of points, where f is not differentiable in $(-3, 3)$ is _____.

Q266. JEE Main 2021 (Online) 25th February Evening Shift

If $\lim\limits_{x\to 0}\frac{ax-(e^{4x}-1)}{ax(e^{4x}-1)}$ exists and is equal to b, then the value of a $-$ 2b is _____.

Q267. JEE Main 2021 (Online) 25th February Morning Shift

The number of points, at which the function
f(x) = | 2x + 1 | $-$ 3| x + 2 | + | x^2 + x $-$ 2 |, x$\in$R is not differentiable, is _____.

Q268. JEE Main 2021 (Online) 24th February Morning Shift

$\lim\limits_{n\to\infty}\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{1+r+r^2}\right)\right\}$ is equal to _____.

Q269. JEE Main 2020 (Online) 6th September Morning Slot

Let f : R $\to$ R be defined as

$f(x)=\begin{cases}x^5\sin\left(\frac{1}{x}\right)+5x^2, & x<0\\ 0, & x=0\\ x^5\cos\left(\frac{1}{x}\right)+\lambda x^2, & x>0\end{cases}$

The value of λ for which f "(0) exists, is _____.

Q270. JEE Main 2020 (Online) 5th September Morning Slot

Let $f(x)=x.\left[\frac{x}{2}\right]$, for -10< x < 10, where [t] denotes the greatest integer function. Then the number of points of discontinuity of f is equal to _____.

Q271. JEE Main 2020 (Online) 4th September Morning Slot

Suppose a differentiable function f(x) satisfies the identity
f(x+y) = f(x) + f(y) + $xy^2 + x^2y$, for all real x and y.
$\lim_{x\to 0} \frac{f(x)}{x} = 1$, then f'(3) is equal to _____.

Q272. JEE Main 2020 (Online) 3rd September Morning Slot

If $\lim_{x\to 0} \left\{ \frac{1}{x^8} \left(1 - \cos\frac{x^2}{2} - \cos\frac{x^2}{4} + \cos\frac{x^2}{2}\cos\frac{x^2}{4}\right)\right\}$ = 2^{-k}

then the value of k is _____ .

Q273. JEE Main 2020 (Online) 2nd September Morning Slot

If $\lim_{x\to 1} \frac{x+x^2+x^3+...+x^n-n}{x-1}$ = 820,
(n $\in$ N) then the value of n is equal to _____.

Q274. JEE Main 2020 (Online) 7th January Evening Slot

If the function ƒ defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by

f(x) = $\begin{cases} \frac{1}{x}\log_e\left(\frac{1+3x}{1-2x}\right), & when\, x \neq 0 \\ k, & when\, x = 0 \end{cases}$

is continuous, then k is equal to_____.

Q275. JEE Main 2020 (Online) 7th January Morning Slot

Let S be the set of points where the function, ƒ(x) = |2-|x-3||, x $\in$ R is not differentiable. Then $\sum_{x\in S} f(f(x))$ is equal to_____.

Q276. JEE Main 2020 (Online) 7th January Morning Slot

$\lim_{x\to 2} \frac{3^x+3^{3-x}-12}{3^{-x/2}-3^{1-x}}$ is equal to_____.

Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **B**	3. **C**	4. **B**	5. **C**
6. **B**	7. **A**	8. **D**	9. **C**	10. **C**
11. **A**	12. **D**	13. **B**	14. **C**	15. **D**
16. **C**	17. **D**	18. **C**	19. **C**	20. **C**
21. **C**	22. **C**	23. **B**	24. **D**	25. **D**
26. **D**	27. **A**	28. **C**	29. **D**	30. **D**
31. **B**	32. **B**	33. **C**	34. **A**	35. **D**
36. **D**	37. **D**	38. **C**	39. **C**	40. **C**
41. **D**	42. **B**	43. **D**	44. **D**	45. **A**
46. **B**	47. **D**	48. **C**	49. **A**	50. **C**
51. **D**	52. **C**	53. **A**	54. **B**	55. **D**
56. **C**	57. **D**	58. **B**	59. **D**	60. **B**
61. **D**	62. **A**	63. **C**	64. **B**	65. **B**
66. **D**	67. **B**	68. **A**	69. **D**	70. **C**
71. **B**	72. **B**	73. **B**	74. **C**	75. **B**
76. **A**	77. **D**	78. **A**	79. **C**	80. **D**
81. **B**	82. **C**	83. **C**	84. **A**	85. **C**
86. **B**	87. **D**	88. **D**	89. **A**	90. **A**
91. **C**	92. **D**	93. **C**	94. **A**	95. **C**
96. **B**	97. **C**	98. **A**	99. **A**	100. **C**
101. **B**	102. **C**	103. **D**	104. **A**	105. **A**
106. **B**	107. **D**	108. **D**	109. **D**	110. **B**
111. **B**	112. **C**	113. **D**	114. **A**	115. **A**
116. **C**	117. **A**	118. **A**	119. **C**	120. **A**
121. **B**	122. **A**	123. **A**	124. **B**	125. **C**
126. **D**	127. **C**	128. **A**	129. **D**	130. **B**
131. **B**	132. **D**	133. **C**	134. **A**	135. **D**
136. **A**	137. **D**	138. **C**	139. **D**	140. **A**
141. **B**	142. **C**	143. **B**	144. **C**	145. **A**
146. **B**	147. **B**	148. **D**	149. **C**	150. **D**

151. **A**	152. **D**	153. **A**	154. **A**	155. **D**
156. **D**	157. **B**	158. **C**	159. **D**	160. **C**
161. **B**	162. **C**	163. **B**	164. **A**	165. **D**
166. **B**	167. **C**	168. **B**	169. **D**	170. **B**
171. **C**	172. **C**	173. **A**	174. **A**	175. **C**
176. **B**	177. **A**	178. **A**	179. **A**	180. **B**
181. **C**	182. **B**	183. **A**	184. **A**	185. **C**
186. **B**	187. **D**	188. **A**	189. **C**	190. **D**
191. **B**	192. **D**	193. **B**	194. **C**	195. **A**
196. **B**	197. **C**	198. **A**	199. **C**	200. **B**
201. **B**	202. **C**	203. **D**	204. **B**	205. **D**
206. **C**	207. **D**	208. **A**	209. **C**	210. **D**
211. **A**	212. **B**	213. **D**	214. **C**	

Numerical

215. **3**	216. **4**	217. **3**	218. **10**	219. **2**
220. **98**	221. **8**	222. **3**	223. **32**	224. **24**
225. **1**	226. **5**	227. **34**	228. **81**	229. **6**
230. **55**	231. **17**	232. **170**	233. **24**	234. **100**
235. **18**	236. **81**	237. **15**	238. **2**	239. **2**
240. **4**	241. **309**	242. **25**	243. **79**	244. **2**
245. **1**	246. **3**	247. **14**	248. **11**	249. **3**
250. **62**	251. **7**	252. **7**	253. **2**	254. **14**
255. **5**	256. **39**	257. **4**	258. **1**	259. **3**
260. **3**	261. **3**	262. **6**	263. **1**	264. **4**
265. **5**	266. **5**	267. **2**	268. **1**	269. **5**
270. **8**	271. **10**	272. **8**	273. **40**	274. **5**
275. **3**	276. **36**			